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Acetylcholine demixes heterogeneous dopamine signals for learning and moving

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Midbrain dopamine neurons promote reinforcement learning and movement vigor. An outstanding question is how dopamine-recipient neurons in the striatum parse these heterogeneous signals. Previous work suggests that cholinergic striatal interneurons may gate dopamine-dependent plasticity, but this has not been tested in behaving animals. Here we studied rats performing a decision-making task with reward-related and movement-related events. Optical measurement of dopamine and acetylcholine release in the dorsomedial striatum (DMS) revealed that reward cues evoked cholinergic pauses with different phase relationships relative to dopamine. When dopamine lagged cholinergic dips, dopamine predicted future behavior and DMS firing rates on subsequent trials. In contrast, when dopamine preceded cholinergic dips, there was no observable relationship between dopamine and learning. Finally, when dopamine was coincident with cholinergic bursts, it preceded and predicted the vigor of contralateral orienting movements. Our findings suggest that cholinergic dynamics determine whether dopamine promotes vigor or learning, depending on the instantaneous behavioral context.

Midbrain dopamine neurons and their terminals in the striatum are implicated in reinforcement learning and motor control[1–11]. Some evidence suggests that different dopamine neurons may be engaged in learning and movement, such that heterogeneous functions could derive from distinct groups of neurons[4,10,12–15]. However, single dopamine neuron axon terminals span millimeters of neuropil[16], and heterogeneous dopamine signals have been observed at single recording sites in the dorsal striatum[4,10,17], suggesting that, for medium spiny neurons, the principal cells of the striatum, dopamine receptor binding reflects multiplexed signals for learning and moving. Here we show that acetylcholine release exhibits distinct dynamics when dopamine encodes reward prediction errors (RPEs) or predicts upcoming movement vigor in rats performing a goal-directed decision-making task.

In reinforcement learning, agents learn the value of taking actions in different states; state-action values are iteratively updated based on RPEs, or the difference between received and expected rewards[18]. A wealth of evidence suggests that dopaminergic neurons instantiate RPEs in the brain[1–8]. One important target of dopamine neurons is the dorsomedial striatum (DMS), a striatal region necessary for learning associations between actions and outcomes[19]. Dopamine is thought to drive plasticity at converging synaptic inputs onto medium spiny neurons such that animals will be more (or less) likely to take actions in states that produced positive (or negative) RPEs.

The dopamine system is also implicated in motor control[20]. Degeneration of dopamine neurons projecting to the dorsal striatum is a hallmark of Parkinson's disease[21]. Lesions or optogenetic perturbations of dopamine neurons or their axon terminals in the DMS produce gross effects on movement initiation, kinematics and vigor[9,11]. Studies have found dopamine responses that reflect movements and cannot be easily reconciled with the RPE framework[4,10,17,22].

Here we tested the hypothesis that striatal acetylcholine dynamically determines whether dopamine drives learning or movement vigor in rats performing a decision-making task. We developed a task that involves several sequential events on each trial; some events are associated with reward-predictive cues, while others elicit orienting movements, which require the DMS[23]. Dopamine reflected RPEs or predicted the vigor of upcoming movements at distinct time points, and dips in acetylcholine occurred at RPE but not movement

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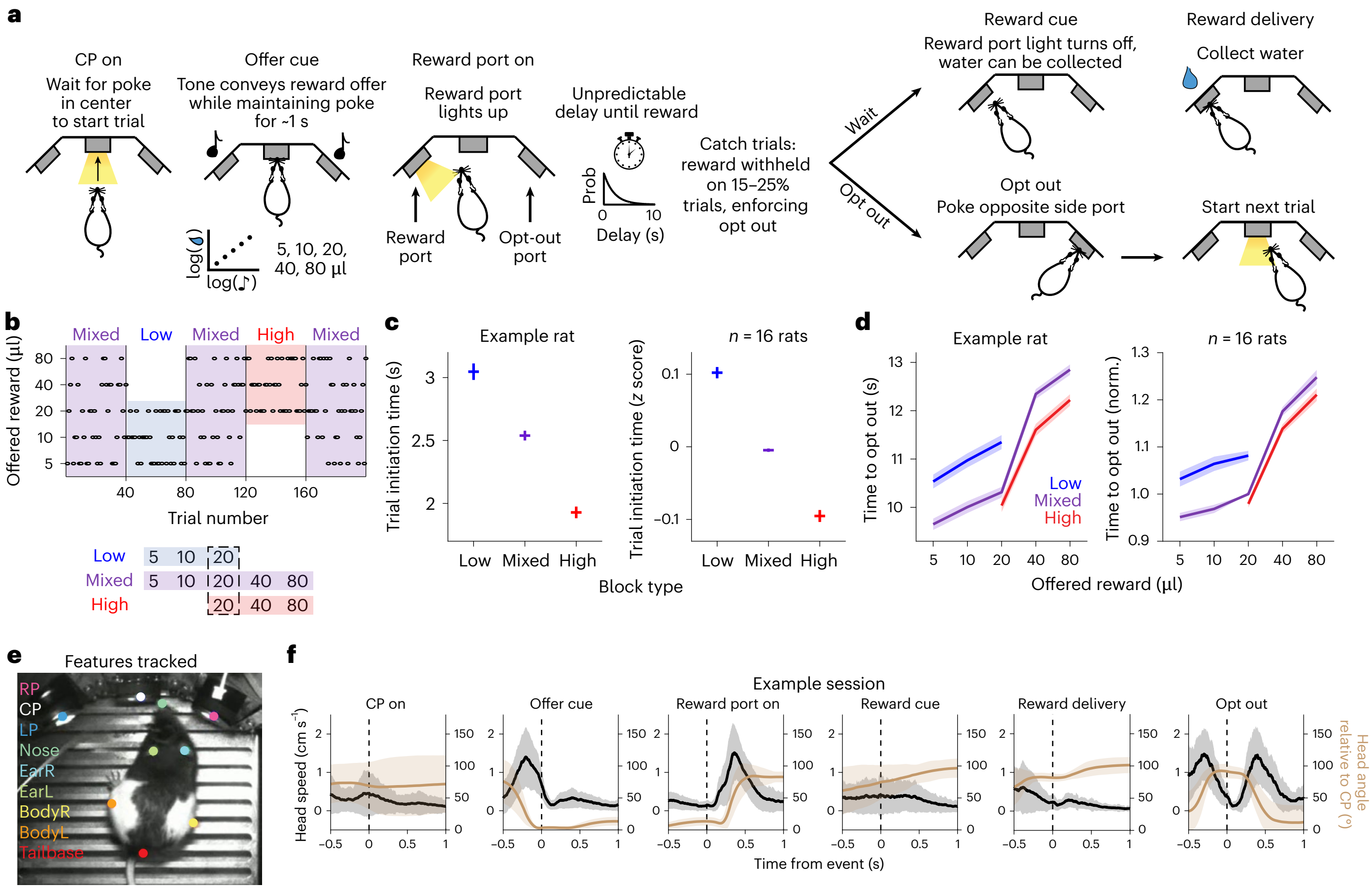

Fig. 1 | Temporal wagering task provides behavioral measures of rats' reward expectations and movement. a, Schematic of behavioral paradigm. **b**, Schematic of the block structure of the task. All sessions start with a mixed block and alternate between low and high blocks. Blocks are 40 completed trials. **c**, Left: average time to initiate trial by block type for an example rat. Trials are pooled across sessions ($n = 63$ sessions, mean ± s.e.m.). Right: z-scored trial initiation time by block, averaged across rats ($n = 16$ rats, mean ± s.e.m.). **d**, Left: average time to opt out by offered reward volume in each block for the same example rat in **c**. Trials are pooled across sessions ($n = 63$ sessions, mean ± s.e.m.). Right: normalized time to opt out by offered reward in each block, averaged across rats ($n = 16$ rats, mean ± s.e.m.). Time to opt out is divided by the time for 20 μl in mixed block for each rat before averaging. **e**, Features tracked with DeepLabCut using videography. **f**, Head speed (black) and angle relative to the central nose port (brown) aligned to different task events in an example session (mean ± 1 s.d. across trials).

events. Dopamine RPEs predicted subsequent behavior when they slightly lagged cholinergic dips, but not when they slightly preceded cholinergic dips. In contrast, we observed bursts in acetylcholine release at contralateral orienting movements. These data suggest that acetylcholine dynamics gate dopamine and striatal function on a moment-by-moment basis.

Results

Temporal wagering task provides measures of rats' reward expectations and movement

We trained rats on a self-paced temporal wagering task that manipulated reward expectations by varying the magnitude of offered rewards in blocks of trials[24] (Fig. 1a,b). Rats initiated trials by poking their nose in the central nose port. This triggered an auditory cue, the frequency of which indicated the volume of water reward offered on that trial (5, 10, 20, 40 or 80 μl). Rats were required to maintain fixation in the central nose poke for ~1 s while the auditory cue played; thus, leaving the nose port early triggered a white noise sound and terminated the trial. At cue offset, the reward was assigned randomly to one of the two side ports, indicated by a light-emitting diode (LED). The rat could wait for an uncued and unpredictable delay to obtain reward, or at any time could opt out by poking in the other side port to start a new trial. On 15–25% of trials, rewards were withheld to assess how long rats were willing to wait for them (catch trials), providing a behavioral measure of their valuation of the offer.

To manipulate reward expectations, rats experienced uncued blocks of trials in which they were presented with low (5, 10 or 20 μl) or high (20, 40 or 80 μl) volumes of water rewards. These were interleaved with mixed blocks that offered all rewards (Fig. 1b). Rats initiated trials more quickly in high blocks compared to low blocks, indicating their subjective value of the environment (Fig. 1c; $n = 16$ rats). We also found that rats modulated how long they were willing to wait for rewards based on the block (Fig. 1d; $n = 16$ rats). In previous work, we found that distinct value computations and neural circuits support the decision to initiate a trial versus opt out during the delay period—initiation times reflect a delta rule and are impacted by perturbing dopamine in the striatum[24–26], while the decision to opt out reflects inference of the current block, and relies on the orbitofrontal cortex[24,27]. Given our previous findings that initiation times rely on a dopamine-dependent reinforcement learning algorithm[24–26], we first focused on that aspect of behavior.

Deep pose estimation[28] showed that rats made orienting movements at certain task events (Fig. 1e,f). When the side LED lit up, indicating that trial's reward port, rats oriented toward it and, after opting out, they oriented back to the center port to initiate a new trial. Thus, reward cues are separable in time from orienting movements, which

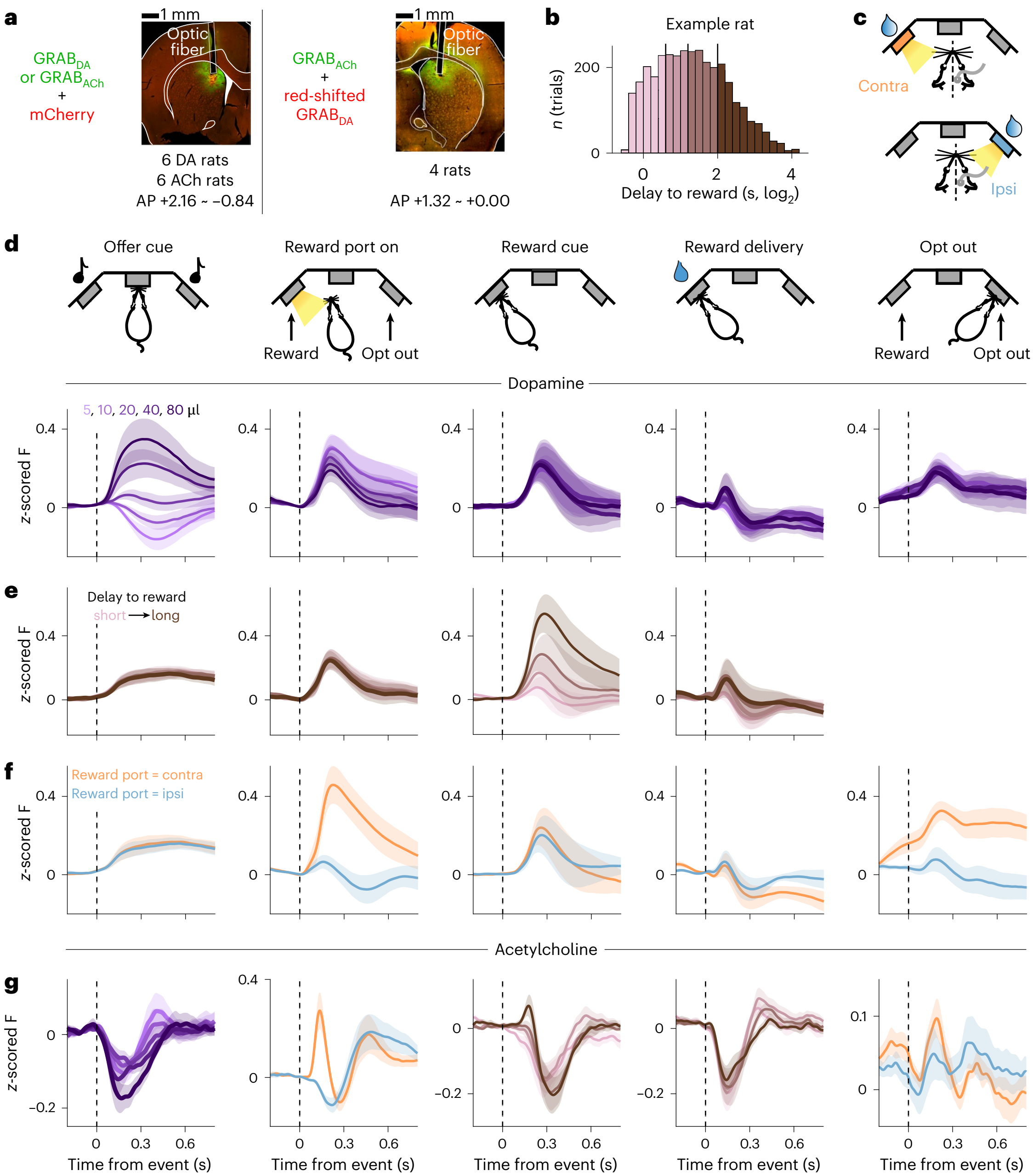

Fig. 2 | Dopamine and acetylcholine in the DMS show distinct dynamics at reward-associated and movement-associated events. a, Left: viral strategy to measure dopamine ($n = 6$ rats, pAAV-hsyn-DA2h) or acetylcholine ($n = 6$ rats, AAV9-hSyn-ACh4.3) release in the DMS using green-fluorescence GRAB sensors and representative histology. Right: viral strategy to measure dopamine (red, pAAV9-hsyn-rDA3m) and acetylcholine (green, AAV9-hSyn-ACh4.3) simultaneously and representative histology ($n = 4$ rats). **b**, Distribution of delay to reward, drawn from an exponential distribution, for an example rat. Colors indicate quartiles. **c**, Schematic of contralateral (orange) and ipsilateral (blue) trials shown for a rat implanted in the right hemisphere. The side of implant is counterbalanced across rats. **d**, Average event-aligned dopamine release during mixed blocks for different reward offers ($n = 10$ rats, mean ± s.e.m.). **e**, Average event-aligned dopamine release split by delay to reward quartiles across all blocks ($n = 10$ rats, mean ± s.e.m.). **f**, Average event-aligned dopamine release split by contralateral (orange) and ipsilateral (blue) trials ($n = 10$ rats, mean ± s.e.m.). **g**, Average event-aligned acetylcholine release split by the task variables most strongly encoded by dopamine at each task event ($n = 10$ rats, mean ± s.e.m.). **d**–**g**, z-scored fluorescence signals are baseline-corrected before pooling across rats. Contra, contralateral; ipsi, ipsilateral; F, fluorescence.

require the DMS[23]. This enables relating dopamine release to multiple reward-predictive cues and event-aligned movements.

Dopamine and acetylcholine show distinct dynamics at reward-associated and movement-associated events

We measured dopamine and acetylcholine release in the DMS in this task. To measure dopamine, we used fiber photometry of an optical dopamine sensor, GRAB$_{DA}$ (Fig. 2a; $n = 6$ rats with pAAV-hsyn-GRAB$_{DA2h}$, $n = 4$ rats with pAAV9-hsyn-GRAB$_{rDA3m}$). We used mCherry or isosbestic control to correct for brain motion[29] in rats injected with DA2h and rDA3m, respectively, with similar results for both methods (Extended Data Fig. 1a,b).

At the offer cue, we observed phasic dopamine release that scaled with the offered reward volume, consistent with an RPE (Fig. 2d). Additionally, because the timing of the rewards was unpredictable and those were omitted on catch trials, we reasoned that, when the rat

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learns reward is available, there should be delay-dependent prediction errors[30,31]. Indeed, dopamine release at the reward cue scaled with delay, again consistent with an RPE (Fig. 2b,e). These RPEs were not modulated by the spatial location of the reward port (Extended Data Fig. 1c,d).

However, at task events that elicited stereotypical orienting movements, dopamine release in the DMS showed a larger phasic response for orienting movements contralateral to the recording hemisphere[4] (Fig. 2c,f). This was true when the reward port was assigned and rats oriented toward that port, and, after opt-outs, when rats oriented back toward the center port to initiate a new trial. Contralateral orienting signals were strongest in the DMS compared to the ventral and dorsolateral striatum (DLS), perhaps reflecting a privileged role for the DMS in executing orienting movements[23] (Extended Data Fig. 2). Dopamine in the ventral striatum showed similar encoding of RPEs as the DMS[4] (Extended Data Fig. 2a–e). Dopamine in the DLS, by contrast, did not exhibit RPE encoding at the offer cue (Extended Data Fig. 2f–h). These data are consistent with the DMS representing an intermediate position along with the continuum of 'reward' to 'action' coding in the striatum and its associated dopaminergic innervation[4,10,15,17]. Computing the discriminability index of dopamine responses for different task variables confirmed that DMS dopamine encoded reward volume, delay and contralateral movements at distinct task events (Methods; Extended Data Fig. 3a–d). Therefore, dopamine in the DMS multiplexed RPEs and contralateral movement signals at distinct time points.

How might dopamine-recipient medium spiny neurons demultiplex or demix these heterogeneous signals? To test the hypothesis that acetylcholine may contribute to this process, we measured DMS acetylcholine release using fiber photometry of the $GRAB_{ACh}$ sensor (Fig. 2a; $n = 10$ rats injected with AAV9-hSyn-ACh4.3). We found prominent dips in acetylcholine release at cues that elicited dopamine RPEs (Fig. 2g). The magnitude of the dip scaled modestly with the offered reward volume at the time of the offer cue. In contrast, when the side LED indicated the reward port, we observed a strong burst in acetylcholine release on trials in which the port was contralateral to the recording site (Fig. 2g), when a phasic increase in dopamine release was also observed (Fig. 2f). At opt out, we also observed a similar but weaker rise in acetylcholine release on contralateral trials (Fig. 2g). The difference in the strength of the acetylcholine burst at the two movement events could reflect more temporal jitter between when rats opt out and when rats orient to initiate subsequent trials (Extended Data Fig. 3e,f). Therefore, acetylcholine exhibits dips or bursts when dopamine encodes RPEs versus reflects contralateral movements, respectively. This pattern was observed when both acetylcholine and dopamine were measured in different animals, as well as when we performed dual-color imaging of $GRAB_{ACh}$ and red-shifted $GRAB_{DA}$ at the same DMS recording site (Extended Data Fig. 4). This shows that differences in cholinergic and dopaminergic dynamics were not due to differences in recording locations. Recording sites spanned ~3 mm across the anteroposterior (AP) axis, so the results appear generalizable across the DMS (see Extended Data Figs. 5–7 for individual histology and dynamics across the AP axis).

Dopamine predicts learning when it lags cholinergic dips

We sought to determine whether dopamine at putative RPE events implemented reinforcement learning. Notably, the relative phase relationship between the cholinergic dips and the dopamine RPEs was different at the offer cue, when dopamine scaled with reward volume, and at the reward cue, when dopamine reflected the reward delay. The cross-correlogram between cholinergic dips and dopamine at the offer cue showed that dopamine lagged acetylcholine by ~100 ms at this time point, and the time to the peak (for dopamine) or trough (for acetylcholine) was significantly different (Fig. 3a; $n = 10$ DA rats, $n = 10$ ACh rats; two-sided Wilcoxon rank-sum test, $P < 0.001$). For visualization, we highlight the relationship between large positive RPEs and cholinergic dips, in which dopamine peaks slightly followed cholinergic dips (the temporal correlation was similar for small volumes that produced negative RPEs, but the sign of the correlogram was flipped; Extended Data Fig. 8a–c). At the reward cue, however, the dopamine peak preceded the cholinergic dip by ~50 ms (Fig. 3b; $n = 10$ DA rats, $n = 10$ ACh rats; two-sided Wilcoxon rank-sum test, $P < 0.001$). Examination of event-aligned responses showed that the dopamine RPE was aligned to the reward cue (when the light turned off after the delay, indicating that reward was baited), but the cholinergic dip was aligned to reward delivery (when the rat poked and triggered the solenoid; Fig. 2e,g). These events occur closely in time, but asynchronously, as the rat has to poke in the nose port after the cue to get reward. Therefore, while dopamine appeared to encode an RPE at both the offer and the reward cue[25,26], its relative phase relationship with cholinergic dips was different at these events.

Phase relationships varied across striatal subregions. In contrast to DMS, dopamine and acetylcholine in the DLS exhibited an antiphase relationship at reward delivery (Extended Data Fig. 8d). The phase relationship may vary depending on the behavioral engagement of striatal subregions.

We previously found that trial initiation times rely on a dopamine-dependent reinforcement learning algorithm[24–26], and are causally impacted by dopamine perturbations[26], so we then sought to relate RPEs at the offer and reward cue to initiation time behavior. We modeled rats' trial initiation times as a linearly decreasing function of the value of the environment, which was iteratively updated by the equation $V_{t+1} = V_t + \alpha\delta$, where δ was an RPE. To evaluate the RPEs at offer cue and reward cue separately, we created two models, in which the RPE reflected either the reward volume ($R_t - V_t$) or the reward delay. Because we observed large RPEs after long delays (Fig. 2e), the delay model RPE scaled with delay ($\mathrm{Delay}_t - V_t$). We simulated the behavior of both models using parameters that minimized the difference between the average model predictions and rat's initiation times as a function of block or previous delay (Fig. 3c,f; Methods). The reward volume RPE model could reproduce the rats' block-dependent initiation times (Fig. 3c). In contrast, because the delay distribution was fixed across blocks, the delay RPE model could not account for changes in initiation times over blocks (Fig. 3f, left). The delay RPE model also predicted faster initiation times after long delays, which was not consistent with rats' behavior (Fig. 3f, right).

The reward volume RPE model predicted that RPEs should scale with offered rewards, which were observed at the time of the offer cue (Fig. 2d). The model also predicted that RPEs should be inversely related to expectations, or the average reward in each block. Indeed, the offer cue dopamine response to 20 μl was largest in low blocks and negative in high blocks[25,26], consistent with a volume-dependent RPE ($n = 10$ rats; two-sided Wilcoxon signed-rank test—low versus mixed, $P = 0.0039$; low versus high, $P = 0.0020$; mixed versus high, $P = 0.0273$). The model predicted that RPEs and behavior would be influenced by previous rewards, with the most recent rewards having the greatest influence and decaying exponentially further in the past (Fig. 3d). Regressing initiation times and dopamine area under the curve (AUC) at the offer cue against previous rewards revealed negative, exponentially decaying influences of previous rewards on dopamine release (Fig. 3d, left) and behavior (Fig. 3d, right). Finally, there was a positive relationship between the magnitude of the phasic dopamine signal at the offer cue and the change in initiation times on the subsequent trial (Methods; Fig. 3e; $n = 10$ rats; linear mixed-effects model, slope = 0.2690, one-sided $P < 0.001$).

In contrast, dopamine at the reward cue reflected the current reward delay, but not past reward delays, which is inconsistent with an iterative reinforcement learning algorithm (Fig. 3g, left). If large RPEs at the reward cue increased the value of the environment, rats would exhibit faster initiation times after long delays (that is, regressing initiation time against delay should yield negative coefficients). However, the previous delay had a small positive coefficient (Fig. 3g, right). Finally, there was no relationship between the magnitude of the phasic

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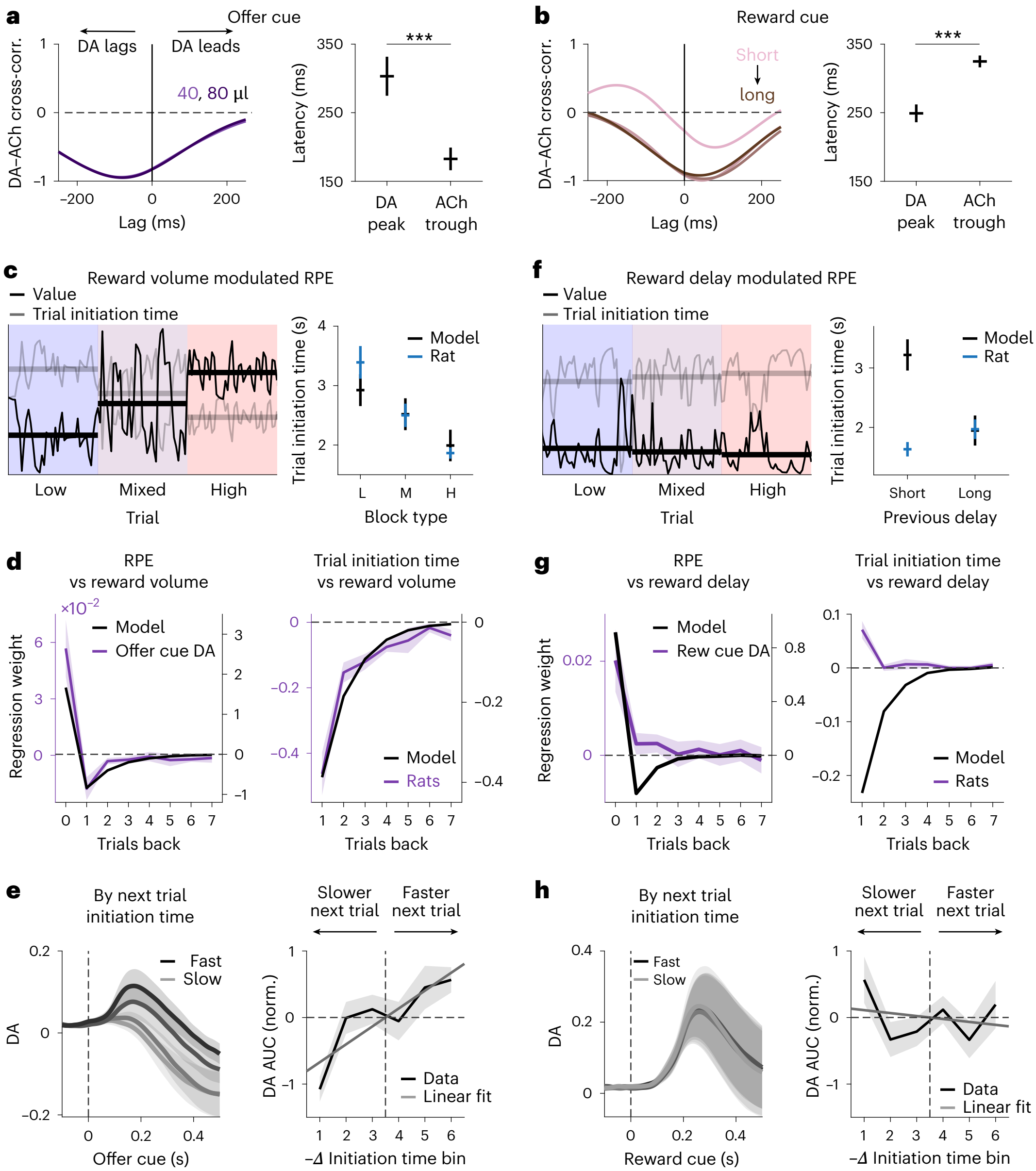

Fig. 3 | Dopamine correlates with learning when it lags cholinergic dips. **a**, Left: cross-correlation of rat-averaged dopamine and acetylcholine signals at the offer cue (−0.1 to 0.5 s) on 40 and 80-µl offer trials in mixed blocks. Negative lag indicates dopamine lags the acetylcholine signal. Right: latency to peak/trough in DA/ACh after the onset of the offer cue on 40 and 80-µl offer trials in mixed blocks (n = 10 DA rats, n = 10 ACh rats, mean ± s.e.m.; two-sided Wilcoxon rank-sum test, $P = 7.20 \times 10^{-4}$). **b**, Left: same as in **a** but at the reward cue on trials with different reward delay quartiles. Right: same as in **a** but after the onset of the reward cue on trials with different reward delay quartiles (n = 10 DA rats, n = 10 ACh rats, mean ± s.e.m.; two-sided Wilcoxon rank-sum test, $P = 1.39 \times 10^{-5}$). **c**, Left: model-predicted value (black) and simulated trial initiation time (gray) for an example rat using reward volume modulated RPE. Solid lines indicate average block values. Right: trial initiation time by block type averaged across rats (blue) and model simulation using trial statistics from the same rats (black; n = 10 rats, mean ± s.e.m.). **d**, Left: dopamine AUC (purple) at the offer cue (0–0.5 s) and model-predicted RPE (black) regressed against reward offers in the last 15 trials. Right: z-scored rat trial initiation time (purple) and model simulation (black) regressed against reward offers in the last 15 trials. Across panels, only mixed blocks are included and seven trials are shown for visualization purposes (n = 10 rats, mean ± s.e.m.). **e**, Left: z-scored dopamine at the offer cue conditioned on trial initiation time on the subsequent trial, averaged across rats (n = 10 rats, mean ± s.e.m.). Dark to light colors indicate the fastest to slowest quartile of initiation time distribution. Right: dopamine AUC at the offer cue (0–0.5 s) versus −Δ trial initiation time, averaged across rats (n = 10, mean ± s.e.m., linear mixed-effects model, slope = 0.2690, one-sided $P < 0.001$). **f**–**h**, Same as in **c**–**e** but using reward delay modulated RPE in model simulation. All blocks are used (**f**); n = 10 rats, mean ± s.e.m., linear mixed-effects model, slope = −0.0442, one-sided $P = 0.7385$ (**h**). For **a** and **b**, ***$P < 0.001$. L, low; M, mixed; H, high; corr., correlation.

dopamine signal at the reward cue and the change in initiation times on the subsequent trial (Methods; Fig. 3h; n = 10; linear mixed-effects model, slope = −0.0442, one-sided $P = 0.7385$).

Therefore, when dopamine slightly lags cholinergic dips at the offer cue, it predicts subsequent initiation times, consistent with acting as an RPE that updates the value of the environment. In contrast, when RPEs slightly precede cholinergic dips, we did not find evidence that they functioned as RPEs to update environmental value for initiation times.

Dopamine fails to predict learning when it precedes cholinergic dips

We then sought to determine what might be learned from the reward cue RPE, which scaled with reward delay. Because we did not observe a strong relationship between reward delay and trial initiation times, we then examined how long rats were willing to wait before opting out of the trial. We have previously modeled this aspect of behavior with a time-varying function that reflects the value of waiting[24,27]. The starting point reflects the reward offer, and the shape reflects the probability of reward over time, dictated by the catch probability, P, and the time constant of the exponential delay distribution, τ (Fig. 4a). The model opts out when the value of waiting falls below the value of the environment, implemented as one of three block-specific thresholds (Fig. 4b). The model captures various aspects of opt out decisions, including sensitivity to blocks and reward offers (Fig. 1d) and dynamics of behavioral changes at various block transitions[24,27].

Manipulating the catch probability P changed how long rats were willing to wait (Fig. 4c; $n = 53$ rats; two-sided Wilcoxon signed-rank test, $P < 0.001$ for all reward offers), consistent with the model[24] (Fig. 4d), suggesting that rats learned this time-varying function through experience. We thus considered the possibility that the delay RPE might update rats' estimates of the value of waiting, that is, temporal reward expectations during the delay. According to temporal difference learning (and belief-state variants[31]), large RPEs after a long delay should increase the value of the states preceding the reward, which in our task should cause rats to wait longer on the subsequent trial (Fig. 4e). However, there was no difference in how long rats waited for rewards in mixed blocks, conditioned on whether the reward delay on the previous trial was in the top versus bottom delay quartile (Fig. 4f). Regressing the time to opt out against past reward delays yielded negligible coefficients (Fig. 4g). In contrast, how long rats waited in mixed blocks before opting out showed a strong relationship with the current reward offer (Fig. 4g; $n = 16$ rats; t test, $P < 0.001$).

We also measured the probability that rats were poking their nose in the reward port during the delay period. This probability exponentially decayed over time in line with the reward hazard rate, defined by P and τ (Fig. 4h). Notably, there was no difference in poking after reward delays in the top versus bottom delay quartiles (Fig. 4i, left). We also compared the latency to first poke in the reward port, which could be influenced by temporal reward expectations. Again, there was no difference in initial poke latencies after long or short reward delays (Fig. 4i, right; $n = 16$ rats; two-sided Wilcoxon signed-rank test, $P = 0.3011$).

Therefore, when dopamine encoded a delay-dependent RPE that slightly preceded cholinergic dips at the reward cue, we failed to find a relationship between dopamine and behavior on subsequent trials, including rats' trial initiation times, time to opt out, probability of poking in the reward port during the delay or initial latency to poke in the reward port. While it is possible that this RPE promoted learning that we did not resolve, this negative result contrasts with dynamics at the offer cue, where dopamine slightly lagged cholinergic dips. At that time point, dopamine predicted trial-by-trial changes in initiation times. These data suggest that the precise phase relationship between cholinergic dips and dopamine RPEs gates whether those RPEs actually promote learning.

Evidence for in vivo plasticity

Because phasic dopamine at the time of the offer cue is consistent with an RPE, we hypothesized that it produces plasticity in the recipient DMS cells[32–34]. Dopamine receptor binding can influence the excitability of medium spiny neurons[35,36], or modify the strength of convergent synaptic inputs[32–34]. However, it has remained unclear how and when RPE-like updating of DMS activity actually occurs in vivo during behavior. For RPE updating to be useful, it must be rapidly occurring but also enduring, and it is not clear how the timescales of conventional short-term and long-term plasticity allow behaviorally useful changes to occur in this manner.

We recorded single-unit action potentials using Neuropixels probes implanted in the DMS (Fig. 5b; $n = 8$ rats). Geometric constraints for chronic implants precluded simultaneous photometry and Neuropixels recordings at the same site. Therefore, we used the trial-by-trial change in initiation times as a behavioral proxy for RPEs on each trial (Fig. 3e). We observed trial-by-trial changes in firing rate (FR) following increases or decreases in initiation times at the time of the offer cue, but not at subsequent events (Fig. 5c). To quantify this effect, we regressed trial-by-trial changes in average FRs for 300 ms after the onset of each event against putative RPEs ($-\Delta$ initiation times; Methods). We used a 300-ms window because dopamine paired with spiking in this time window produces plasticity in medium spiny neurons[37]. The highest proportion of cells had significant slope parameters at the time of the offer cue (51.6%, 130/252 units) with 53.1% (69/130 units) having a significant positive slope (for example, example cell in Fig. 5d) and 46.9% (61/130 units) having a significant negative slope (Fig. 5e,f). These groups may reflect neurons expressing D1 receptors and D2 receptors, respectively. The relationship between putative RPEs and trial-by-trial changes in FR decreased at events that were further from the offer cue, suggesting that plasticity was temporally specific to the time of the RPE. We confirmed that the results in Fig. 5c–f reflect RPEs from the previous trial rather than the initiation time on the current trial (Extended Data Fig. 9a,b).

To characterize the timescale of these effects, we identified sequences of trials in which a large positive/negative RPE estimated from behavior (top and bottom 15%) on trial n was followed by a negligible RPE ($|\mathrm{RPE}| < 0.2$) on trial $n + 1$. If plasticity effects are long-lasting, the change in FR from trial n to trial $n + 1$ should persist to trial $n + 2$, as there was a negligible RPE on the intervening trial. Figure 5g shows the FRs of two example neurons on such a trial sequence. We analyzed FRs for all such sequences for cells with positive and negative FR correlations with RPEs. In both groups, positive RPEs on trial n changed FRs on trial $n + 1$, and that change persisted to trial $n + 2$ (Fig. 5h; two-sided Wilcoxon signed-rank test—positively modulated cells, trial n versus $n + 1$, $P = 0.0004$; trial n versus $n + 2$, $P = 0.0065$; trial $n + 1$ versus $n + 2$, $P = 0.4169$; negatively modulated cells, trial n versus $n + 1$, $P < 0.001$; trial n versus $n + 2$, $P < 0.001$; trial $n + 1$ versus $n + 2$, $P = 0.1898$). These data are consistent with RPEs producing changes in task-evoked FRs that endured over trials. This result held across a range of thresholds that are used to select RPEs on trial n (Extended Data Fig. 9c). Negative RPEs produced more variable effects on trial-by-trial spiking for these specific sequences of trials (Extended Data Fig. 9d), possibly reflecting different reliability of plasticity mechanisms following increases versus decreases of dopamine. Nonetheless, positive RPEs in the DMS modulate evoked firing on subsequent trials at the time of the offer cue but not at neighboring task events, including the reward cue (Fig. 5c–e), and those changes persist over trials (Fig. 5g,h).

Dopamine predicts movement vigor when coincident with cholinergic bursts

Next, we sought to characterize the contralateral selective dopamine signal at events accompanying orienting movements, when dopamine was coincident with cholinergic bursts (Fig. 6a,b). The time to peak in dopamine signal at these events preceded that in head speed by ~100 ms (Fig. 6c,e; $n = 24$ sessions from five rats; t test—reward port on, $P = 0.0135$; opt out, $P < 0.001$). Furthermore, the amplitude of the dopamine signal at these events predicted the vigor of the orienting movement. When the reward port was assigned by the LED, rats oriented toward that port. The AUC of the dopamine signal at this event was significantly larger when reaction times were in the fastest quartile compared to the slowest quartile (Fig. 6d; $n = 10$ rats; two-sided Wilcoxon signed-rank test, black dots indicate time points with $P < 0.05$). This was true when we restricted trials to those with the same offer volume (Extended Data Fig. 10c; $n = 10$ rats; two-sided Wilcoxon signed-rank test, $P = 0.0137$). Notably, neither movement vigor (Extended Data Fig. 10a) nor dopamine release (Extended Data Fig. 10b) at this time point were positively correlated with the reward offer volume, consistent with this orienting movement reflecting a prepotent

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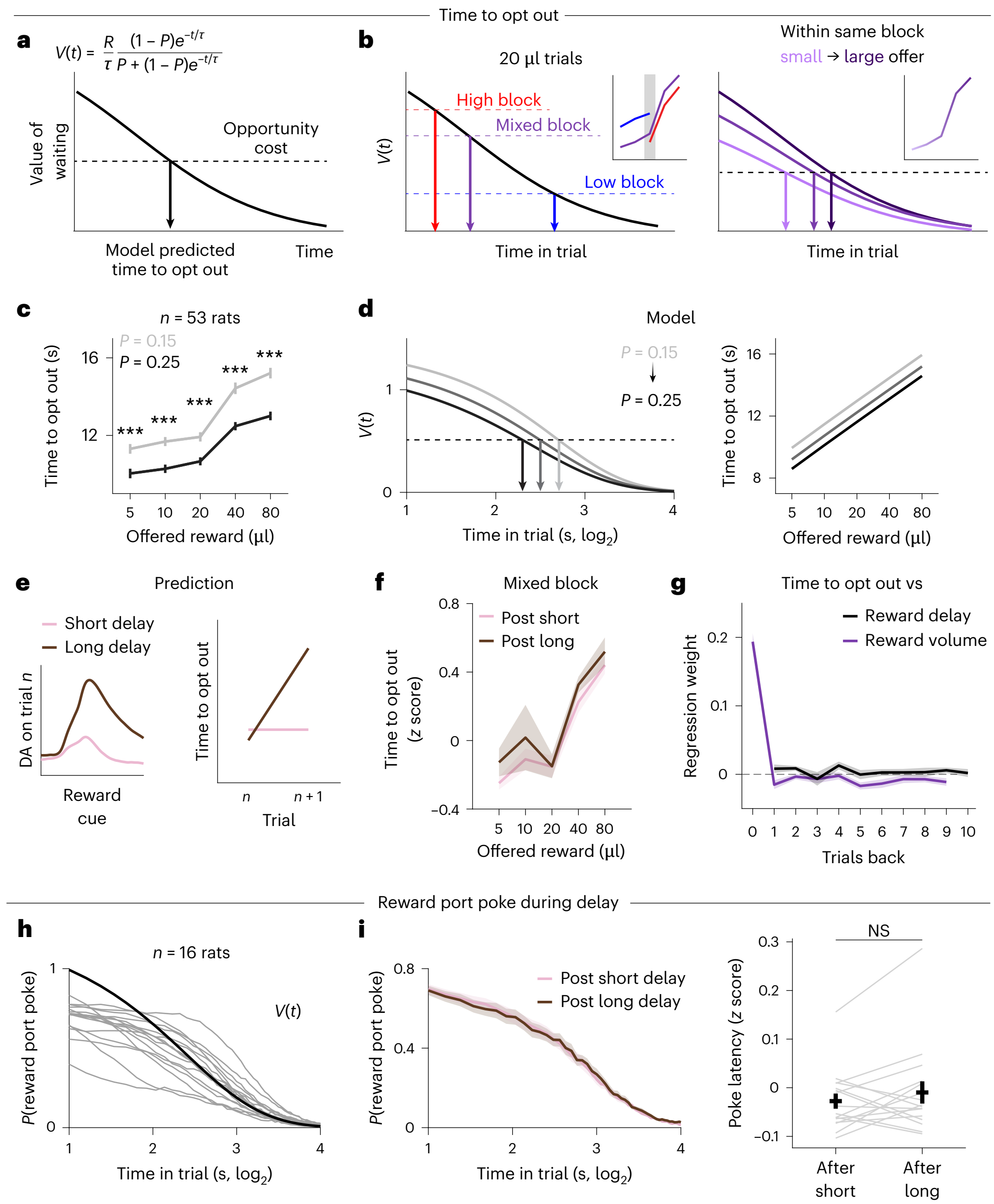

Fig. 4 | Dopamine fails to predict learning at reward cue when it precedes cholinergic dips. a, Value of waiting during the delay period on a given trial as a function of P(catch) (probability of the trial being unrewarded) and τ (time constant of the exponential delay distribution). The model opts out when the value of waiting falls below the opportunity cost, which is fixed for a given block. **b**, Left: block-specific opportunity cost predicts different time to opt out on 20-µl offer trials, consistent with rats' behavior. Right: in a given block, value function predicts monotonic relationship between reward offer and time to opt out, consistent with rats' behavior. Insets show rats' behavior from Fig. 1d. **c**, Average time to opt out in mixed blocks for rats tested with catch probabilities of 0.15 (light gray) and 0.25 (black; $n = 53$ rats, mean ± s.e.m., two-sided Wilcoxon signed-rank test—5 µl, $P = 2.13 \times 10^{-6}$; 10 µl, $P = 4.62 \times 10^{-7}$; 20 µl, $P = 6.98 \times 10^{-7}$; 40 µl, $P = 1.58 \times 10^{-8}$; 80 µl, $P = 2.11 \times 10^{-9}$). **d**, Value function when P(catch) changes (left) and model-predicted time to opt out in mixed blocks for different values of P(catch) (right). **e**, Large RPEs after long delay trials (brown) should increase the value of preceding states and thus the time to opt out on the subsequent trial. Negligible DA evoked on short delay trials (pink) predicts little or no change in time to opt out on the subsequent trial. **f**, z-scored time to opt out during mixed blocks conditioned on reward delay on the previous trial, averaged across rats ($n = 16$ rats, mean ± s.e.m.). Short (pink) and long (brown) delays are defined as the bottom/top quartile of the reward delay distribution across sessions for each rat. **g**, Time to opt out regressed against reward delay (black) and reward volume (purple) in the past 15 trials during mixed blocks, averaged across rats ($n = 16$ rats, mean ± s.e.m., t test—time to opt out versus reward delay, trial back = 2, $P = 0.0223$; trial back = 4, $P = 0.0398$; time to opt out versus reward volume, trial back = 0, $P < 0.001$; trial back = 1, $P = 0.0344$; trial back = 5, $P = 0.0046$; trial back = 6, $P = 0.0473$). Only ten trials are shown for visualization purposes. **h**, Probability of nose poking in the reward port during the delay period in mixed blocks overlaid with task-defined value function ($\tau = 2.5$, P(catch) = 0.25). Gray lines show individual rats ($n = 16$ rats). **i**, Left: probability of nose poking in the reward port during the delay period after trials in the bottom (pink) and top (brown) quartiles of reward delay distribution ($n = 16$ rats, mean ± s.e.m.). Right: z-scored latency to poke in the reward port after the reward port light turns on after short versus long reward delay trials, averaged across rats ($n = 16$ rats, mean ± s.e.m., two-sided Wilcoxon signed-rank test, $P = 0.3011$).

response to a salient stimulus[38]. Therefore, variability in movement speed and dopamine release was independent of reward expectations at this event. Similarly, after rats opt out, they orient back to the central nose port to start a new trial, and the AUC of the dopamine signal at opt out was also larger when rats were faster to initiate the next trial (Fig. 6f; $n = 10$ rats; two-sided Wilcoxon signed-rank test, black dots indicate time points with $P < 0.05$). Therefore, the phasic dopamine signal at these movement-associated events preceded and predicted the vigor of upcoming contralateral orienting movements.

To test whether dopamine at this event promoted plasticity, we tested for a significant correlation between trial-by-trial changes in the FRs of DMS cells and dopamine release (using the reaction time on the previous contralateral trial as a proxy for dopamine). In contrast to RPE events, when behavioral proxies for dopamine predicted trial-by-trial FRs, we found that, for the overwhelming majority of cells, there was no significant correlation (Extended Data Fig. 9e,f). However, we found that DMS cells encoded movement vigor at this time point: we trained a logistic regression decoder to predict whether the reaction time on each trial was in the fastest or slowest quartile, based on the FRs of simultaneously recorded neurons (Methods; Fig. 6g). Cross-validated classification accuracy was high and above chance in 17 of 18 sessions (Fig. 6h), suggesting that dopamine and DMS spiking at movement-associated events predicted the vigor of contralateral orienting movements but did not result in persistent changes in task-evoked spiking of DMS cells.

Discussion

Although cholinergic interneurons only comprise 1–3% neurons in the striatum[39], they are thought to be critical for basal ganglia function. Tonically active neurons (putative cholinergic interneurons) develop pauses in firing after cues that predict appetitive or aversive outcomes[40–46], suggesting that cholinergic pauses support learning[47,48], possibly through local modulation of dopamine release[49–51]. Our data suggest that the relative phase relationship between dopamine and acetylcholine[52–54] determines dopamine's action on recipient neurons. Recent work suggests that temporal dynamics of dopamine release can vary depending on the behavioral engagement of striatal subregions[55]. We speculate that modulation of the phase relationship between dopamine and acetylcholine provides a circuit mechanism for arbitration across behavioral controllers originating from different striatal subregions; when striatal subregions do not support behavior, or dopamine-dependent plasticity in those subregions is not desirable, dopamine RPEs may lead (instead of lag) cholinergic dips. Moreover, dopamine signals that seem inconsistent with RPE (for instance, phasic bursts to cues that predict aversive outcomes[56]) may be more interpretable in the context of concurrent cholinergic dynamics.

Dopamine neurons exhibit heterogeneous responses during complex tasks[10,15], and in some cases different response profiles map onto transcriptional cell types[57–59]. We find that, in the DMS, dopamine RPE and movement-predictive signals are present at the same recording site. While these signals may reflect inputs from different dopamine neurons, if axonal arbors overlap[16], medium spiny neurons will receive heterogeneous dopamine inputs.

The temporal relationship between dopamine and acetylcholine release that is important for medium spiny neurons likely reflects several complex factors, including receptor affinity, efficacy and dynamics of reuptake, and the timescale of interactions between second messengers downstream of dopaminergic and cholinergic G protein-coupled receptors. Here we showed that dopamine predicted movement vigor or subsequent behavior depending on its phase relationship with

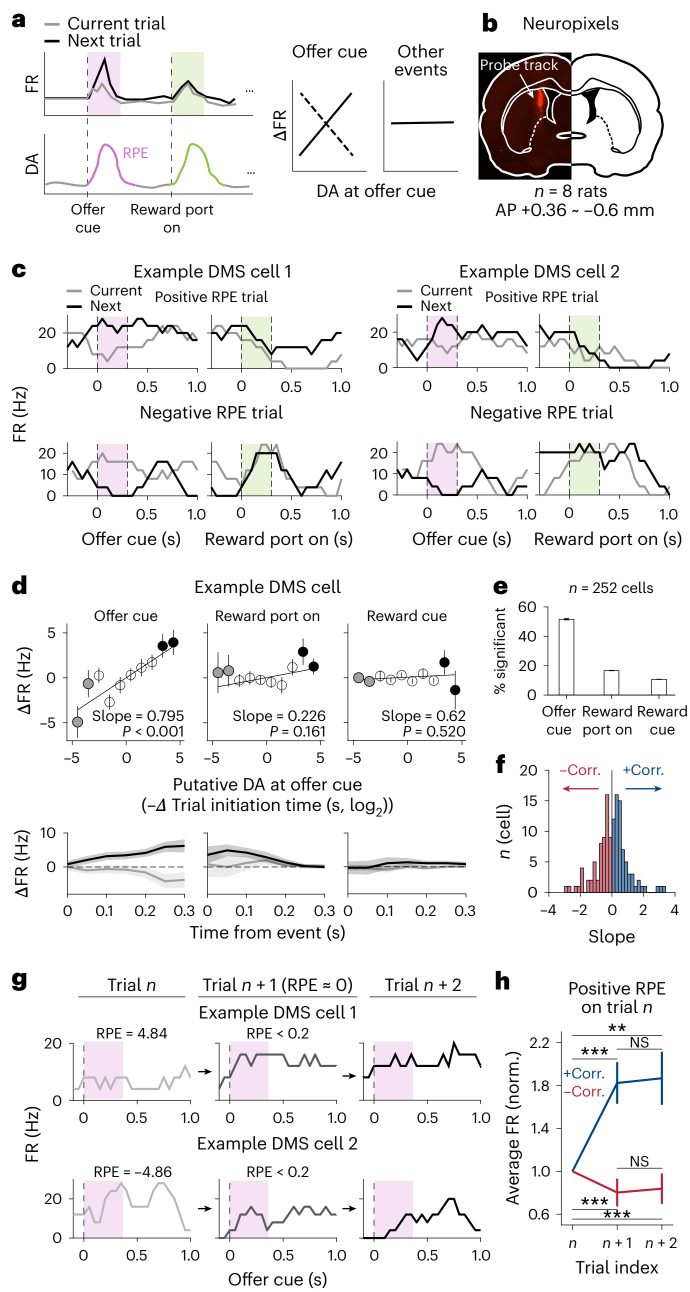

Fig. 5 | Behavioral proxy for dopamine predicts DMS FR on subsequent trials. **a**, Schematic of the prediction for state-specific or input-specific dopamine-dependent synaptic plasticity where RPE at the time of the offer cue produces trial-by-trial changes in FRs at that specific event but not at other states. **b**, Example histology of Neuropixels probe track in the DMS. **c**, FR of two example cells on current (gray) and the subsequent trial (black) after a positive and negative RPE. Shaded boxes indicate the time window used to compute average FR (0–0.3 s) in the following panels. **d**, Top: trial-by-trial changes in average FR surrounding each task event (0–0.3 s) versus behavioral proxy for RPE ($-\Delta\log_2$ trial initiation time) for an example cell. Solid lines show the best-fit line (two-sided t test on slope—offer cue, $P < 0.001$; reward port on, $P = 0.161$; reward cue, $P = 0.520$) and the filled black/gray circles indicate trials with large positive/negative RPEs, respectively. Trials are binned in 15 linearly spaced intervals for visualization (mean ± s.e.m.). Bottom: change in average FR on trials with large positive (black) and negative RPEs (gray) for the same example cell in the top panel (mean ± s.e.m. across trials). **e**, Fraction of cells with significant slopes at different task events. Error bars are binomial confidence intervals. **f**, Histogram of slopes for significantly modulated cells at the offer cue (red, positively modulated by RPE; blue, negatively modulated by RPE). **g**, FR of two example cells aligned to the onset of offer cue on three consecutive trials where a big positive/negative RPE on the first trial is followed by a negligible RPE on the second trial. **h**, Change in average z-scored FR at offer cue (0–0.3 s) of positively (blue) and negatively (red) modulated cells on three consecutive trials for which a large positive RPE (top 15%) on trial n is followed by a negligible RPE on trial $n+1$ ($n = 145/164$ trial sequences for +corr./−corr., mean ± s.e.m., two-sided Wilcoxon signed-rank test—positively modulated cells, trial n versus $n+1$, $P = 0.0004$; trial n versus $n+2$, $P = 0.0065$; trial $n+1$ versus $n+2$, $P = 0.4169$; negatively modulated cells, trial n versus $n+1$, $P = 2.74 \times 10^{-11}$; trial n versus $n+2$, $P = 9.57 \times 10^{-15}$; trial $n+1$ versus $n+2$, $P = 0.1898$). For **h**, $^{**}P < 0.01$, $^{***}P < 0.001$.

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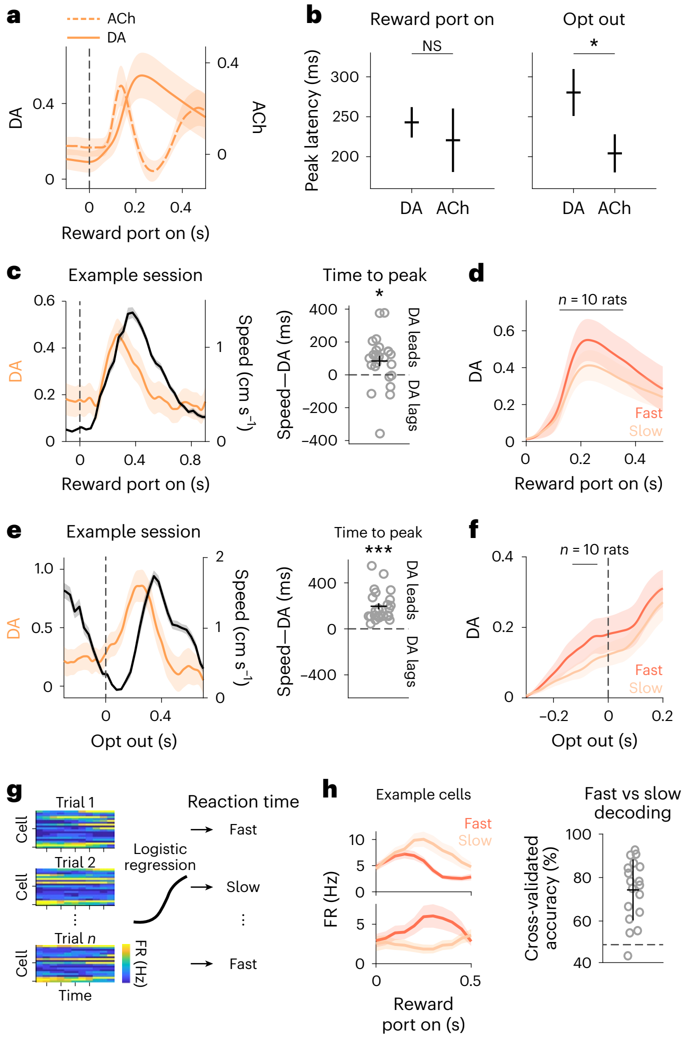

Fig. 6 | Dopamine predicts movement vigor when coincident with cholinergic bursts. a, z-scored DA (solid orange) and ACh (dashed orange) responses on contralateral trials aligned to when the side LED turns on, averaged across rats ($n = 10$ DA rats, $n = 10$ ACh rats, mean ± s.e.m.). **b**, Latency to peak in dopamine and acetylcholine after the side LED turns on (left) and when rats opt out (right) on contralateral trials, averaged across rats ($n = 10$ DA rats, $n = 10$ ACh rats, mean ± s.e.m.; two-sided Wilcoxon rank-sum test—reward port on, $P = 0.1109$; opt out, $P = 0.0483$). **c**, Left: dopamine release on contralateral trials (orange) when the side LED turns on, overlaid with head speed (black) in an example session (mean ± s.e.m. across trials). Right: difference in time to peak in dopamine compared to head speed ($n = 24$ sessions from five rats, mean ± s.e.m. across sessions, two-sided t test, $P = 0.0135$). Empty circles indicate individual sessions. **d**, z-scored dopamine release on contralateral trials aligned to side LED onset, split by reaction time (fast, bottom quartile; slow, top quartile). Reaction time quartiles were defined within each session, and session-averaged responses were computed for each rat before pooling across rats ($n = 10$ rats, mean ± s.e.m.). Black line indicates time points with significant differences between fast and slow trials (two-sided Wilcoxon signed-rank test, $P < 0.05$). **e**, Same as in **c** but for opt out (two-sided t test, $P = 1.39 \times 10^{-7}$). **f**, Same as in **d** but at opt out split by the speed of the following trial initiation (fast, bottom quartile; slow, top quartile; $n = 10$ rats, mean ± s.e.m.). Sessions were aggregated to define trial initiation quartiles for each rat as the number of opt-out trials are limited in each session. **g**, Schematic of a logistic regression decoder that takes population responses of 0–0.5 s at the onset of side LED as input to predict reaction time on a given contralateral trial. **h**, Left: average FR of two simultaneously recorded example cells on fast (dark orange) versus slow (light orange) reaction time trials (mean ± s.e.m. across trials). Right: average cross-validated classification accuracy of a logistic regression decoder. Dashed horizontal line indicates chance performance ($n = 18$ sessions from eight rats, mean ± s.e.m. across sessions). Gray circles indicate individual sessions. For **b**, **c** and **e**, $^*P < 0.05$, $^{***}P < 0.001$.

acetylcholine, but the ground truth temporal dynamics experienced by medium spiny neurons may be different than what we measured. For instance, indicator kinetics presumably impacted the signals we measured. We used both high-affinity ($GRAB_{DA2h}$) and medium-affinity dopamine sensors ($GRAB_{rDA3m}$), which have similar onset kinetics[60]. Indeed, there were no significant differences in timing of positive dopamine RPEs with medium versus high-affinity sensors (Extended Data Fig. 8e), although offset kinetics might be more sensitive to sensor affinity[60].

It is technically challenging to perform causal perturbations to precisely manipulate the phase relationship between acetylcholine and dopamine. Cholinergic interneurons can actively regulate dopamine release by nicotonic and muscarinic receptors on dopamine terminals[49–51], and dopamine can also impact cholinergic dynamics through D2 receptors on cholinergic interneurons[61,62]. Therefore, optogenetically stimulating one neuromodulator will likely impact the dynamics of the other, although the effects on their precise temporal relationship are difficult to predict. This highlights the need for new and better tools for manipulating neuromodulatory signaling in vivo.

While it is unclear why the reward cue RPE did not impact behavior, at least by the metrics we evaluated, we speculate that it may be because knowing the reward delay on trial t does not help one predict the reward delay on trial $t+1$; the delay is inherently stochastic, and its uncertainty is irreducible (that is, it reflects expected uncertainty[25,63]). Therefore, learning from this RPE will not help one maximize future rewards. An interesting future question is how animals might learn that RPEs at different events reflect distinct types of uncertainty[63], and how cholinergic dynamics evolve over that learning process.

Classic studies suggested that corticostriatal plasticity is governed by a three-factor rule requiring coincident presynaptic and postsynaptic spiking and dopamine receptor binding[34,64,65]. Recent work has revised this view, suggesting that a fourth factor—pauses in cholinergic interneurons—is required for induction of plasticity[66]. Our findings support this hypothesis, although the mechanism is unclear. We hypothesize that G-proteins downstream of dopamine and acetylcholine receptors have opposing effects on intracellular second messengers such as adenylyl cyclase. For instance, M4 receptors are found on D1-receptor-expressing medium spiny projection neurons[67]. M4 receptors are coupled to $G_{i/o}$ proteins, and, when activated, inhibit adenylyl cyclase and downregulate cAMP. Consequently, M4 receptor activation can counteract the positive cAMP signaling induced by dopamine binding to D1 receptors. This suggests that, at tonic concentrations of dopamine and acetylcholine, potentiation and depression mechanisms are in competition, with phasic dopamine favoring potentiation through D1-mediated protein kinase A activation, and acetylcholine favoring depression through M4 activation. However, cholinergic pauses may relieve M4-mediated suppression of adenylyl cyclase and allow D1-mediated enhancement of protein kinase A. The mechanism could also be presynaptic—corticostriatal and thalamostriatal afferents express M2 receptors, which suppress glutamatergic transmission, and blocking those receptors promotes long-term potentiation[68–70]. Therefore, cholinergic pauses may relieve M2-mediated suppression of glutamatergic inputs to the striatum[68]. The relative phase relationships that gate or block plasticity likely reflect the impact of receptors with different affinities (for example, D1 is low affinity, M2 and M4 are high affinity) on the relative timing of intracellular second messengers.

Despite the prominence of the dopamine RPE hypothesis, it has remained unclear how and when RPE-like updating of striatal activity actually occurs during behavior. While reinforcement learning models of behavior assume that learning is reflected in trial-by-trial changes in state-action values (Q values), represented in corticostriatal synaptic weights[71], studies in slices typically evaluate effects of long-term plasticity protocols on the order of minutes after tetanic stimulation or prespike/postspike pairing. Our data suggest that induction of temporally specific plasticity (either short term or long term) can occur within the seconds-scale intertrial intervals in our task, while expression of plasticity through changes in striatal FRs is persistent over trials. The temporal specificity of effects at RPE but not movement events further suggests that trial-by-trial changes in FRs did not simply reflect dopaminergic modulation of excitability[35,36]. These results add to recent findings in other brain regions of rapid in vivo plasticity that operates on behavioral timescales[72], and validate key assumptions of reinforcement learning accounts of behavior.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41593-026-02227-x.

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Methods

Rats

A total of 79 Long–Evans rats (60 males, 19 females, 6–24 months of age) were used in this study. Of these, 22 rats underwent fiber photometry, 8 underwent electrophysiology and 53 were included in the behavioral analysis shown in Fig. 4c; 4 rats participated in both fiber photometry and behavioral experiments.

Among the 22 wild-type rats used for fiber photometry, 13 rats were implanted in the DMS only (6 expressing $GRAB_{DA2h}$, 6 expressing $GRAB_{ACh}$ and 1 coexpressing $GRAB_{rDA3m}$ and $GRAB_{ACh}$), 3 rats in the DMS and DLS in opposite hemispheres (all coexpressing $GRAB_{rDA3m}$ and $GRAB_{ACh}$) and 6 rats in the NAcc (all expressing $GRAB_{DA2h}$). For behavioral analysis in Figs. 1 and 4, the 16 DMS rats were used, except in Fig. 4c.

Among the eight rats used for electrophysiology experiment, three rats were TH-Cre transgenic and one rat was Adora2a transgenic; the remaining rats were wild-type. No behavioral difference in task performance was observed between transgenic and wild-type rats.

Rats were water-restricted to motivate task performance. At the end of the day, rats were given 20 min of access to ad libitum water. After behavioral sessions on Friday until mid-day Sunday, rats received ad libitum water. Animal use procedures were approved by the New York University Animal Welfare Committee (2021-1120) and carried out in accordance with National Institute of Health standards.

Behavior training and task logic

Rats were trained in incremental stages of task complexity in a high-throughput behavioral facility using a computerized training protocol as described previously[24]. Briefly, LED illumination from the central nose port indicated that the animal could initiate a trial by poking its nose in that port ('CP on'). Upon trial initiation, the center LED turned off. While in the central nose port, rats were required to maintain fixation for a duration sampled from a uniform distribution between 0.8 and 1.2 s. During the fixation period, a tone played from both speakers, the frequency of which indicated the volume of the offered water reward for that trial (1, 2, 4, 8, 16 kHz, indicating 5, 10, 20, 40, 80-µl rewards; 'offer cue'). After the fixation period, one of the two side LEDs was illuminated, indicating that the reward might be delivered at that port ('reward port on'); the side was randomly chosen on each trial. This event also initiated a variable and unpredictable delay period, which was randomly drawn from an exponential distribution with mean = 2.5 s. The reward port LED remained illuminated for the duration of the delay period, and rats were allowed to freely behave during this period. When reward was available, the reward port LED turned off ('reward cue'), and rats could collect the offered reward by nose poking in that port ('reward delivery'). The rat could also choose to terminate the trial at any time by nose poking in the opposite, unilluminated side port ('opt out'), after which a new trial would immediately begin. On a proportion of trials (15–25%), the delay period would only end if the rat opted out (catch trials). If rats did not opt out within 100 s on catch trials, the trial would terminate.

The trials were self-paced—after receiving their reward or opting out, rats were free to initiate another trial immediately. However, if rats terminated center fixation prematurely, they were penalized with a white noise sound and a time-out penalty, typically lasting 2 s ('violation'). The trial after a premature fixation break offered the rats the same amount of reward to disincentivize premature terminations for small volume offers.

We introduced semi-observable, hidden states in the task by including uncued blocks of trials with varying reward statistics[73]—high and low blocks, which offered the highest three or lowest three rewards, respectively, were interspersed with mixed blocks, which offered all volumes. There was a hierarchical structure to the blocks, such that high and low blocks alternated after mixed blocks (for example, mixed-high-mixed-low or mixed-low-mixed-high). Each session always started with a mixed block. Blocks transitioned after 40 successfully completed trials. Because rats prematurely broke fixation on a subset of trials, in practice, blocks varied in the number of trials.

Rats were considered to have achieved task proficiency when their time to opt out exhibited linear sensitivity to the offered reward volume and time to opt out for the 20-µl offer in low blocks was longer than in high blocks for at least 3 consecutive days. Rats that met these criteria underwent experimental-specific surgeries. After recovering from surgeries, rats were retrained until they achieved the same presurgery performance, which typically took one session. Rats were then habituated to being tethered for 2–3 days before proceeding with recordings and/or manipulations.

Surgery

Rats were anesthetized (isofluorane at 4% for induction and 1.5–2.5% for maintenance) and placed in a stereotaxic frame. Skull screws were inserted to secure implants before proceeding with surgeries specific to experiments.

Fiber photometry. For fiber photometry experiment with separate dopamine (n = 6 rats) and acetylcholine (n = 6 rats) recordings, rats were injected with 300–480 nl of either $GRAB_{DA2h}$ (AAV9-hSyn-$GRAB_{DA2h}$; Addgene, 140554) or $GRAB_{ACh}$ (AAV9-hSyn-ACh4.3, WZ Biosciences, YL10002) into the DMS (AP = −0.4 mm, mediolateral (ML) = ±3.5 mm, dorsoventral (DV) = −4.6 mm or AP = 1.7 mm, ML = ±2.8 mm, DV = −4.3 mm from the skull surface at bregma at a 10° angle). To allow for brain motion correction, mCherry (AAV9-CB7-CI-mCherry-WPRE-RBG; Addgene, 105544) was injected as a control fluorophore by mixing with the GRAB sensors at a ratio of 1.5:1–2.5:1 before injection. One rat injected with $GRAB_{ACh}$ did not receive mCherry and an isosbestic signal was used as the control.

For fiber photometry experiments involving simultaneous dopamine and acetylcholine recordings (n = 4 rats), rats were injected with a total of 1,000–1,080 nl of $GRAB_{rDA3m}$ (pAAV9-hsyn-rDA3m; WZ Biosciences, YL002032-AV9) and $GRAB_{ACh}$ (AAV9-hSyn-ACh4.3; WZ Biosciences, YL10002) mixed at a ratio of 1:1–1:1.25.

All rats were then bilaterally implanted with 400-µM core, 0.5 NA optical fibers (Thorlabs) above the injection area. Craniotomies were sealed with Kwiksil and fibers were secured to the skull using Dentin and Metabond. Rats were allowed to recover for a minimum of 7 days before behavioral training resumed.

Electrophysiology. The day before surgery, the Neuropixels probe assembly was prepared by mounting the probe (Neuropixels probe 1.0) onto a custom-designed three-dimensional printed piece and staining it with DiI. OpenEphys built-in self-tests were run on the probe assembly before surgery. Rats were implanted with a probe assembly in the DMS (AP = −0.24 mm, ML = ±2.5 mm, DV = −4.4 mm from the skull surface at bregma; six rats in the left hemisphere, two rats in the right hemisphere). A ground wire was inserted in the cerebellum and fixed with Dentin. Craniotomies were covered with Kwik-Cast and gaps between the probe assembly and the skull were sealed with sterile paralube vet ointment. The probe was then fixed to the skull with Metabond and dental acrylic. Rats were allowed to recover for a minimum of 5 days before behavioral training resumed.

Markerless pose tracking

Videos and synchronization pulses were acquired using methods described below for each experiment type. DeepLabCut software (v2.2.3) was used for tracking the behavior ports (right port, central port and left port) and six body parts (right ear, nose, left ear, midpoint along the right torso, midpoint along the left torso and base of the tail). Frames to be annotated and used as a training set were

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selected using k-means clustering. For each training cycle, 100–200 frames were extracted and annotated. The training was initialized with ResNet-50 and run for 500,000 iterations. Upon completion of training, the model-predicted positions of the features of interest were visually inspected. Outlier frames were then extracted manually and re-annotated. Frame annotation and training were repeated until no significant outliers existed, which typically took four to five training cycles with a total of 400–600 annotated frames.

The tracked features were smoothed in time with a moving median filter of 0.167 s. To obtain head speed and angle relative to the behavior ports for analysis, the centroid of the head was calculated by averaging three body parts on the head (earR, earL and nose). Extreme outliers in head speed and angle were removed by truncating the 99th percentile. Head speed and angle were aligned to task events by aligning the synchronization pulses with Bpod behavior events.

Fiber photometry experiment

Video and fiber photometry data acquisition. Video data were acquired using ceiling-mounted cameras to capture a top-down view of the arena (Doric USB3 behavior camera, Sony IMX290, recorded with Doric Neuroscience Studio (v6) software). To synchronize with neural recordings and Bpod task events, the camera was connected to a digital I/O channel of a fiber photometry console (FPC) and triggered at 30 Hz through a transistor–transistor logic (TTL) pulse train. The remaining three digital I/O channels of the FPC were connected to the three behavior ports to record TTL pulses when port LEDs turned on and off. An analog channel of the FPC was connected to low-autofluorescence patch cords to record fluorescence. Fluorescence from activity-dependent ($GRAB_{DA}$ and $GRAB_{ACh}$) and activity-independent (isosbestic or mCherry) signals was acquired simultaneously through demodulation and downsampled on-the-fly by a factor of 25 to ~481.9 Hz. Excitation wavelengths were 405 nm (isosbestic), 470 nm (green fluorescent protein (GFP)) and 560 nm (red fluorescent protein). For three-channel acquisition used in simultaneous dopamine and acetylcholine recordings, the red fluorescent protein excitation wavelength was 568 nm. The photometry rig included a rotary fluorescence minicube (RFMC6_IE(400-410)_E1(460-490)_F1(500-540)_E2(550-570)_F2(580-680)_S). The power at the tip of the fiber optic was 20–30 µW for each excitation channel.

Fiber photometry data preprocessing. The recorded demodulated fluorescence was corrected for photobleaching and motion using Two-channel Motion Artifact Correction (TMAC)[29] with mCherry or isosbestic signal as the activity-independent channel. TMAC uses a Bayesian inference-based algorithm to infer motion artifacts from the control channel (that is, an activity-independent indicator). For the cohort of rats where dopamine and acetylcholine were recorded separately, nine of ten rats were injected with mCherry. For the cohort injected with $GRAB_{ACh}$ and $GRAB_{rDA3m}$ for simultaneous recording, isosbestic signal of $GRAB_{ACh}$ was used as the control channel. The excitation wavelength we used for the isosbestic point is based on previously characterized properties of the $GRAB_{ACh3.0}$ sensor, and what has been used as the isosbestic point for this sensor in past studies[74]. The motion-corrected signal was z scored and then aligned to task events using synchronization TTL pulses from the behavior ports. It is possible that the $GRAB_{ACh4.3}$ sensor we used has a slightly shifted isosbestic point from the $GRAB_{ACh3.0}$ sensor[74]. However, the fact that results were qualitatively the same regardless of motion correction with mCherry compared to the isosbestic point shows that this did not impact the results or conclusions of our study (Extended Data Fig. 4).

Electrophysiology experiment

Neuropixels data acquisition. Spike data were acquired from 384 channels at 30 kHz using Neuropix-PXI hardware and OpenEphys. To synchronize with behavior, task event TTL pulses were recorded simultaneously using NI-DAQmx.

Spike sorting. Spike data were preprocessed using Kilosort2.5. After preprocessing, clusters that were identified from Kilosort as single units were manually inspected using the Phy2 Python package. Clusters with unreliable waveform shape, autocorrelogram or spike amplitude time course were manually classified as multi-unit or noise and removed from further analysis. For the identified single units, spike data were aligned to behavioral events and binned into 50-ms bins using custom MATLAB codes.

Event-aligned photometry signals

To align photometry signals to task events, motion-corrected signals were z scored within each session before pooling sessions for each rat. Event-aligned responses were then averaged across rats. For analyses split by reward delay quartiles, trials with reward delays shorter than 0.75 s were excluded to avoid bleed-over responses from the reward port on event. When baseline correction was applied, the minimum value of the condition-averaged signal (for example, different offer volumes, delay quartiles, reward port side) during the baseline window (−0.1 to 0 s for all events except opt out; −0.3 to 0 s for opt out) was subtracted from all trials corresponding to that condition.

Quantification of task variable encoding

To quantify dopamine encoding of task variables for Extended Data Figs. 2e and 3d, we computed the discriminability index (d') of event-aligned dopamine responses, conditioned on the offered reward volume, reward delay and reward port side. Because reward port side is a binary variable (contralateral versus ipsilateral), we grouped the other task variables into two categories for consistency—for reward volume, we compared 5 and 10 µl with 40 and 80-µl offer trials; for reward delay, we compared the bottom 50% versus top 50% reward delay distribution. For each rat, we first computed the average z-scored event-aligned signal for each condition. Then, the discriminability index was calculated as:

$$d' = \frac{\mu_A - \mu_B}{\sqrt{0.5(\sigma_A^2 + \sigma_B^2)}}$$

where μ and σ are the mean and s.d., respectively, of the distribution across sessions.

Behavioral model

We modeled the trial initiation times as a linearly decreasing function of the value of the environment, which was updated according to the following recursive equation: $V_{t+1} = V_t + \alpha\text{RPE}$. The RPE was defined as $R_t - V_t$ for the volume RPE model and $\text{Delay}_t - V_t$ for the delay RPE model. The learning rate $\alpha \in (0,1)$ governed the rate at which value estimates were updated across trials. Trial statistics from the same rats used in behavioral analyses were used to simulate the model. Best-fit parameters for the learning rate α and the scaling factor β, which determined the relationship between value and trial initiation time, were obtained by simulating a grid of parameter combinations and selecting those that qualitatively best captured rats' behavior. The final parameter pairs used were $(\alpha, \beta) = (0.5, 0.48)$ for the volume RPE model and (0.46, 0.54) for the delay RPE model.

Trial initiation versus dopamine AUC analysis

To relate dopamine at the offer cue to trial-by-trial changes in trial initiation time (Fig. 3e), we restricted the analysis to postviolation trials, as prior work showed that reward volume modulation of initiation time is driven primarily by these trials[24]. Specifically, we analyzed how dopamine on a violation trial n related to changes in trial initiation time between trials n and $n+1$.

For each rat, we removed outliers in initiation times by excluding the bottom and top 10% values pooled across all sessions. We then computed the change in trial initiation times as follows: $-\Delta$ initiation = $-$(initiation time$_{n+1}$ $-$ initiation time$_n$). A positive value of $-\Delta$ initiation means that the rat initiated the subsequent trial faster. $-\Delta$ initiation values were then split into six bins—the negative and positive values were each divided into terciles, resulting in the first three bins for trials with subsequently slower initiation and the last three bins for subsequently faster initiation.

To compute dopamine AUC, z-scored photometry signal was aligned to the offer cue and baseline-corrected using the same method described in 'Event-aligned photometry signals'. The AUC was calculated from 0 to 0.5 s after offer cue onset on each trial. Trials with extreme AUC values (bottom and top 0.1%) were excluded. For each rat, the average AUC was computed within each of the six initiation bins and then normalized by z scoring across bins before pooling across rats.

For the reward cue analysis (Fig. 3h), the procedure was identical except that dopamine AUC was aligned to the reward cue, and trial initiation changes were calculated from a given reward cue trial n to trial $n+1$.

Trial-by-trial spiking analysis

To account for biases in statistical test due to imbalanced dataset across task events, we performed random upsampling of the minority dataset. If the ratio of the number of data points between the majority and the minority datasets exceeded 2.5, the minority dataset was upsampled through random sampling with Gaussian noise ($\sigma = 1$). Only sessions that had at least 20 trials were included for analysis.

Logistic regression model

To compare encoding of movement vigor in DMS spikes, we trained a logistic regression decoder. For each session, the predictor tensor matrix (trial × neuron × time) was built by concatenating spikes from 0 to 0.5 s after the onset of reward port assignment. For each trial in a given session, it was classified as fast/slow if it were in the bottom/top quartile of the reaction time distribution. Only sessions with at least five simultaneously recorded DMS units were included for analysis. We used the MATLAB function fitclinear to train the decoder to predict 1 if a given trial is fast, or 0 if slow. To cross-validate, we implemented tenfold cross-validation with ridge regression. The regularization strength was determined by training the decoder sequentially on a set of 11 logarithmically spaced regularization strengths ranging from 10^{-4} to $10^{-0.2}$ and selecting the value that minimizes classification error.

Histology

After completion of each experiment, rats were deeply anesthetized and transcardially perfused with PBS and formalin for histological verification. Brains were postfixed with formalin for 3 days and sectioned at 50-µm coronal slices on a Leica VT1000 vibratome. For electrophysiology experiments, sections were imaged with an Olympus VS120 Virtual Slide Microscope to verify probe location. For fiber photometry experiments, sections were stained to enhance GFP and imaged to verify virus expression and fiber implant location.

Immunohistochemistry. For brain tissues acquired for fiber photometry experiment, sections were treated with 1% hydrogen peroxide in 0.01 M PBS for 30 min, washed with 0.01 M PBS and blocked in 0.01 M PBS containing 0.05% sodium azide and 1% bovine serum albumin (PBS–BSA–azide) for 30 min before treating with primary antibodies. GFP was amplified with rabbit anti-GFP (Thermo Fisher Scientific, A11122; 1:2,000, lot 2083201) as primary antibody and Goat anti-Rabbit IgG Alexa Fluor 488 (Thermo Fisher Scientific, A11008; 1:200, lot 2147635) as secondary antibody. Primary and secondary antibodies were made in PBS–BSA–azide, and primary antibodies additionally contained 0.2% Triton X. Sections were incubated in primary antibody overnight at room temperature, washed with PBS for 15 min and treated with secondary antibody for 1 h in dark. After washing with PBS for 15 min, sections were mounted on glass slides using Prolong Gold (Vector Laboratories).

Statistics and reproducibility

Statistical analyses were performed in MATLAB (R2021a; The MathWorks) using built-in functions. No statistical method was used to predetermine sample size; sample sizes were chosen based on prior studies in the field and practical considerations related to experimental feasibility. The experiments were not randomized, and the investigators were not blinded to allocation during experiments and outcome assessment.

Outlier trial initiation times, defined as values falling within the top or bottom 1% of each rat's cumulative trial initiation time distribution pooled across sessions, were excluded from analysis. These exclusions were applied uniformly unless otherwise noted in the main text or Methods (for example, in 'Trial initiation versus dopamine AUC analysis'). Details for data exclusion regarding photometry and electrophysiology analyses are provided in 'Event-aligned photometry signals' and 'Trial-by-trial spiking analysis'

Reproducibility was assessed by confirming that key effects were observed consistently across animals and sessions, as indicated by reporting exact sample sizes and by plotting individual data points where appropriate.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The data used in this study are deposited on Zenodo at https://doi.org/10.5281/zenodo.17457318 (ref. 75; behavioral data, electrophysiology data, DeepLabCut data), https://doi.org/10.5281/zenodo.17458072 (ref. 76; rat metadata, postprocessed photometry and behavioral data), https://doi.org/10.5281/zenodo.17458437 (ref. 77; DMS photometry data, single-color imaging), https://doi.org/10.5281/zenodo.17460638 (ref. 78; DLS photometry data, dual-color imaging), https://doi.org/10.5281/zenodo.17478653 (ref. 79; DMS photometry data, dual-color imaging) and https://doi.org/10.5281/zenodo.17486595 (ref. 80; NAcc photometry data, single-color imaging).

Code availability

Code used to analyze the data and generate the figures is available at https://github.com/constantinoplelab/published/tree/main/DMS_AChDA.

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Author contributions

H.J.J. and C.M.C. designed the study and wrote the paper. H.J.J., R.M.W. and C.E.M.G. collected the data, and H.J.J. analyzed the data. C.M.C. advised on the data analysis and supervised the project.

Competing interests

All authors declare no competing interests.

Additional information

Extended data is available for this paper at https://doi.org/10.1038/s41593-026-02227-x.

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41593-026-02227-x.

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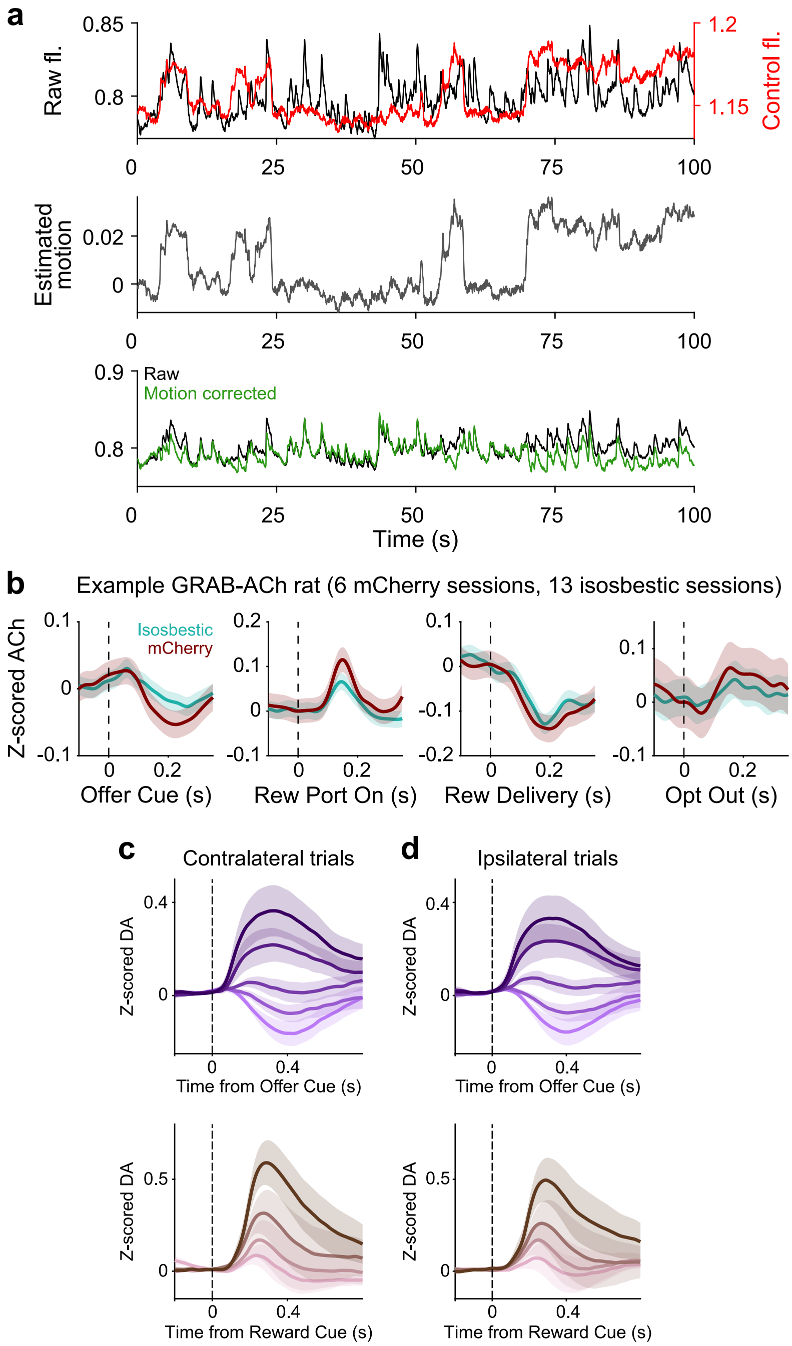

Extended Data Fig. 1 | Motion correction and ipsilateral/contralateral RPEs. **a**, Top, example raw fluorescence (black) overlaid with simultaneously recorded mCherry fluorescence (red). Middle, estimated motion calculated with TMAC (gray). Bottom, raw fluorescence overlaid with TMAC motion-corrected fluorescence (green). **b**, Average motion-corrected acetylcholine signal using isosbestic (teal, N = 13 sessions) or mCherry (red, N = 6 sessions) as the activity-independent channel for an example rat (mean ± s.e.m.). **c**, Z-scored dopamine release split by offered reward volume in mixed blocks at the time of the offer cue (top) and by reward delay quartile at reward cue (bottom) on contralateral trials. **d**, Same as in **a** but on ipsilateral trials. N = 10 rats, mean ± s.e.m. (**c**,**d**).

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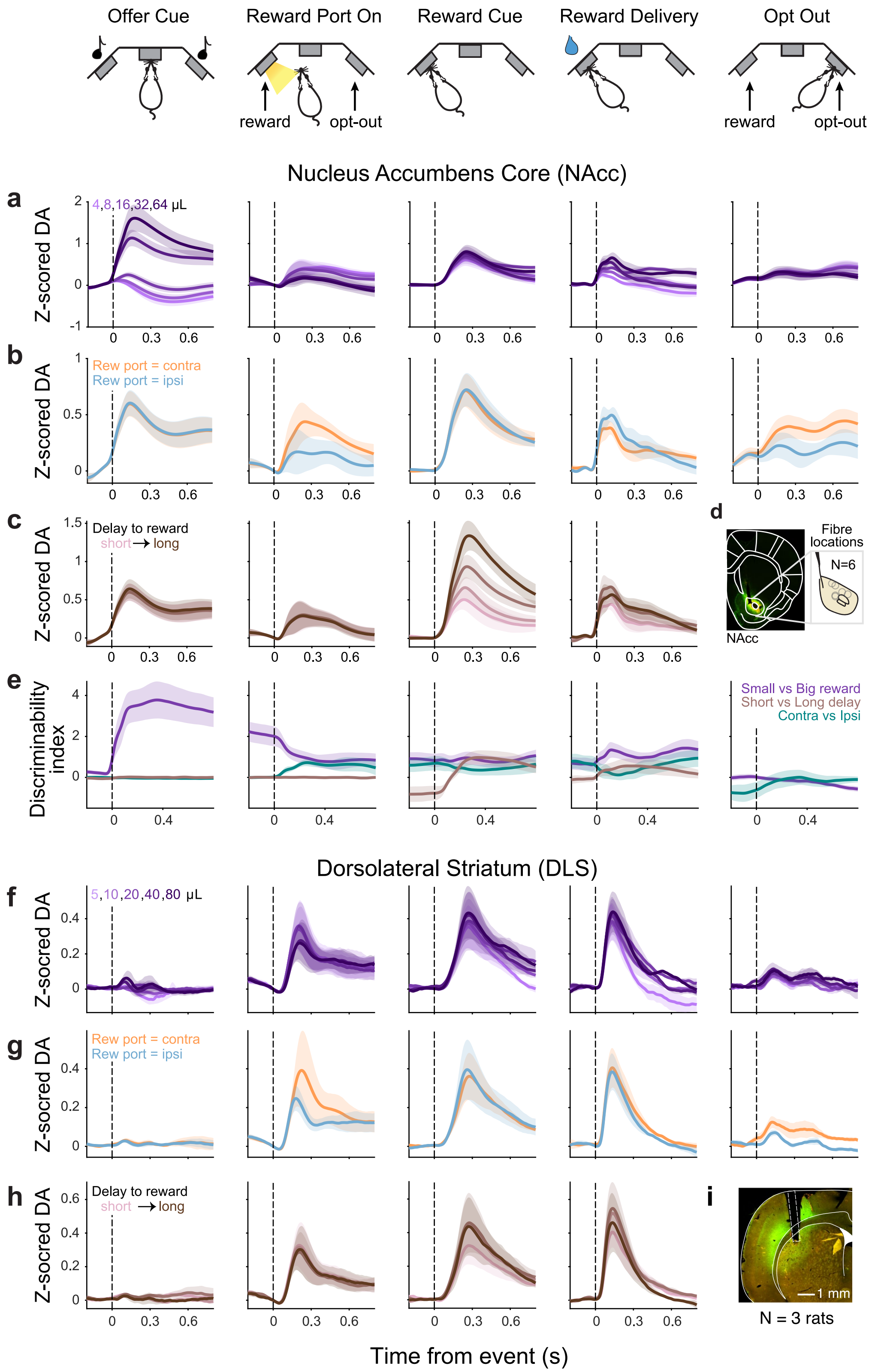

Extended Data Fig. 2 | Dopamine release in the nucleus accumbens core and dorsolateral striatum. a, Average event-aligned dopamine release in the NAcc split by offered reward volume in mixed blocks. These data were collected from female rats, so they were offered slightly smaller water volumes compared to males, to obtain sufficient numbers of behavioral trials[24,26] (4–64 µl versus 5–80 µl for males). **b**, Average event-aligned dopamine release in the NAcc split by contralateral and ipsilateral trials. **c**, Average event-aligned dopamine release in the NAcc split by reward delay quartiles across all blocks. **a**–**c**, Dopamine signals are z-scored and baseline-corrected before pooling across rats. **d**, Representative histology image of viral expression of GRAB$_{DA2h}$ and mCherry and recovered fiber placements. **e**, Average discriminability index of different task variables (purple: reward volume, brown: reward delay, green: contra/ipsi) aligned to all task events. Across **a**–**e**, N = 6 rats, mean ± s.e.m. **f**–**h**, Event-aligned dopamine release in the DLS for different reward offers during mixed blocks (**f**), reward port location (**g**), and delay to reward quartiles (**h**), averaged across rats (N = 3 rats, mean ± s.e.m.). These rats were recorded with medium-affinity red-shifted GRAB sensor (rDA3m). Dopamine signals are z-scored and baseline corrected before pooling across rats. **i**, Representative histology image of viral expression of GRAB$_{rDA3m}$ and GRAB$_{ACh}$ and fiber track in the DLS.

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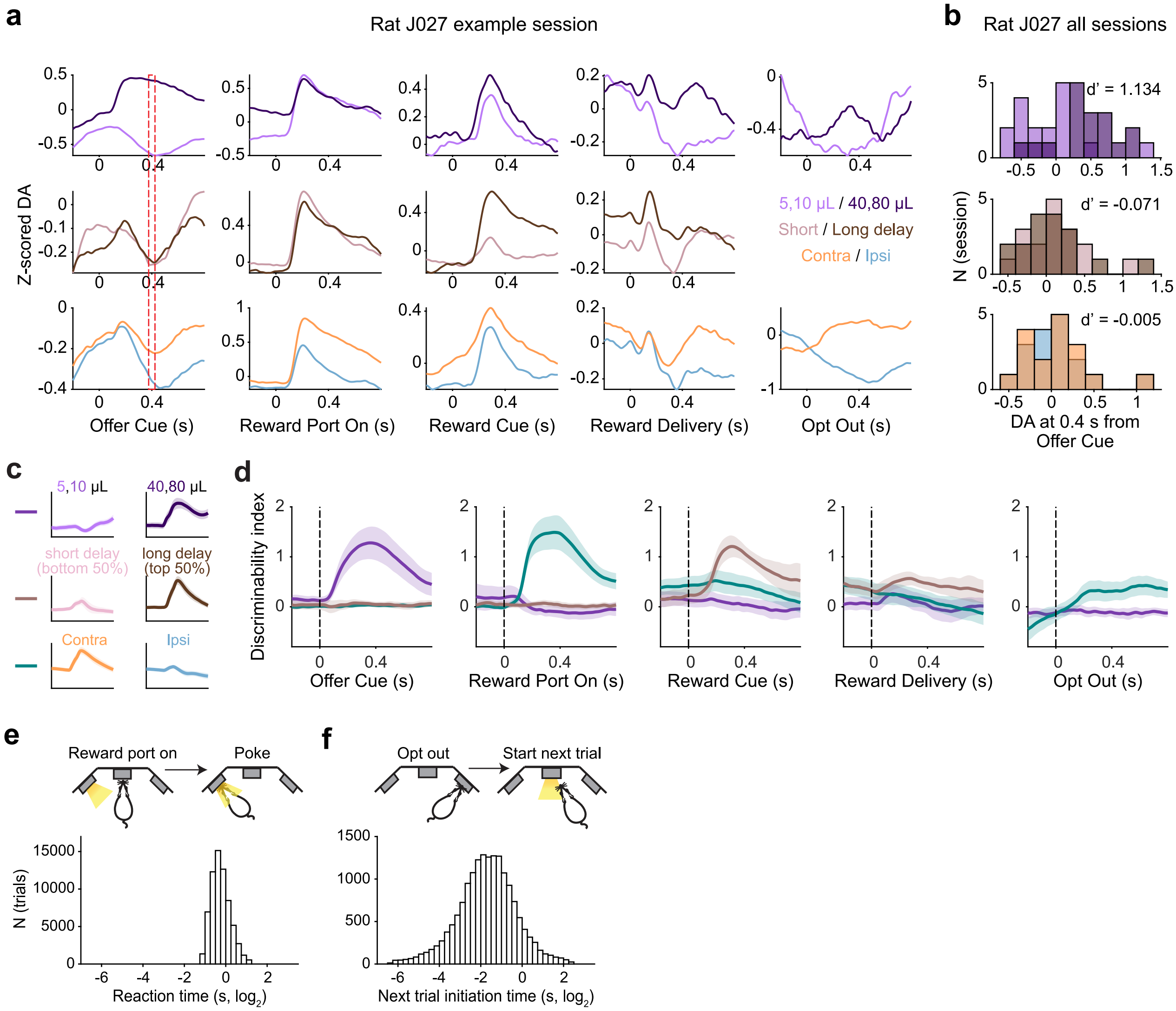

Extended Data Fig. 3 | Discriminability index quantifies encoding of task variables at different events. a, Top, average z-scored dopamine release on small (5, 10 µl) versus large (40, 80 µl) reward offer trials aligned to all task events in an example session. Middle, average z-scored dopamine release on short (bottom 50% of reward delay distribution) versus long (top 50%) reward delay trials. Bottom, average z-scored dopamine release on contralateral versus ipsilateral trials. Across panels, the red vertical line indicates the time point the histograms in **b** were generated from (t = 0.4 s from the offer cue onset). **b**, Distribution of z-scored dopamine at 0.4 s from the offer cue onset (red vertical line in **a**) across 17 sessions from an example rat. **c**, Schematic of how task variables are binarized to compute discriminability index. **d**, Average discriminability index of different task variables (purple: reward volume, brown: reward delay, green: contra/ipsi) aligned to all task events (N = 10 rats, mean ± s.e.m.). **e**, Histogram of rats' reaction time to the side LED lighting up to indicate the side of the reward port. **f**, Histogram of rats' time to initiate trial following opt-outs. N = 16 rats (**e**,**f**).

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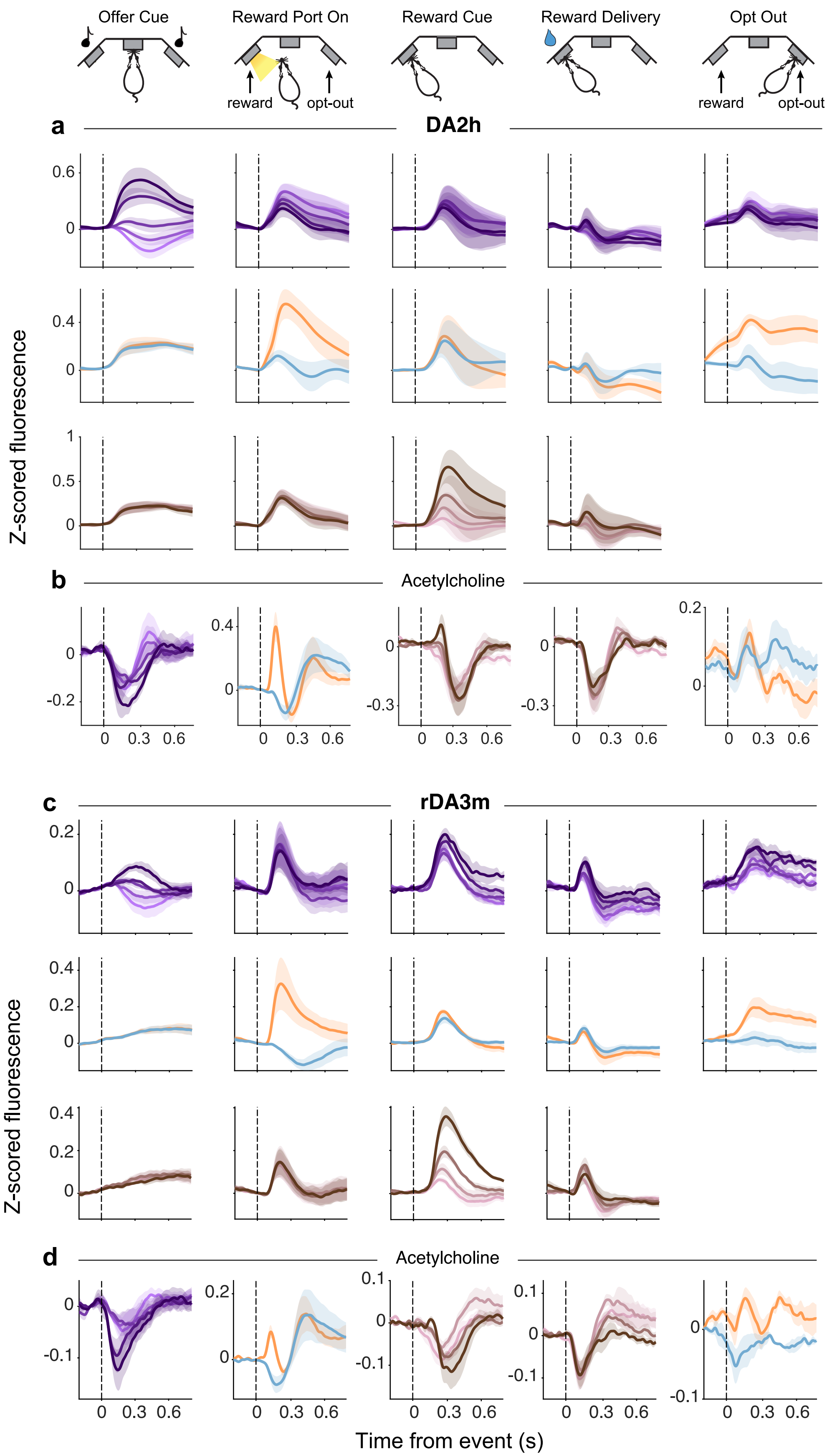

Extended Data Fig. 4 | See next page for caption.

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Extended Data Fig. 4 | Acetylcholine and high-affinity dopamine sensor dynamics in the DMS in different animals, and acetylcholine and medium-affinity dopamine sensor in the DMS at the same recording site show similar dynamics. a, Event-aligned z-scored dopamine signals measured with high-affinity sensor, split by reward offer volume (top, light to dark purple for small to large volumes), reward port location (middle, orange for contralateral and blue for ipsilateral), and delay to reward quartile (bottom, pink to brown for shortest to longest delay quartile bin), averaged across rats (N = 6 rats, mean ± s.e.m.). Across panels, signals are baseline corrected (Methods). **b**, Same as in **a** but for acetylcholine. Recordings in panels **a** and **b** were performed in separate animals. **c**, Event-aligned z-scored dopamine signals measured with medium-affinity sensor, split by reward offer volume (top, light to dark purple for small to large volumes), reward port location (middle, orange for contralateral and blue for ipsilateral), and delay to reward quartile (bottom, pink to brown for shortest to longest delay quartile bin), averaged across rats. **d**, Same as in **c** but for simultaneously measured acetylcholine. Recordings in panels **c** and **d** were performed in the same animals, at the same recording site. **c**,**d**, N = 4 rats, data are mean ± s.e.m. across rats (**c**,**d**). Across panels, signals are baseline corrected (Methods).

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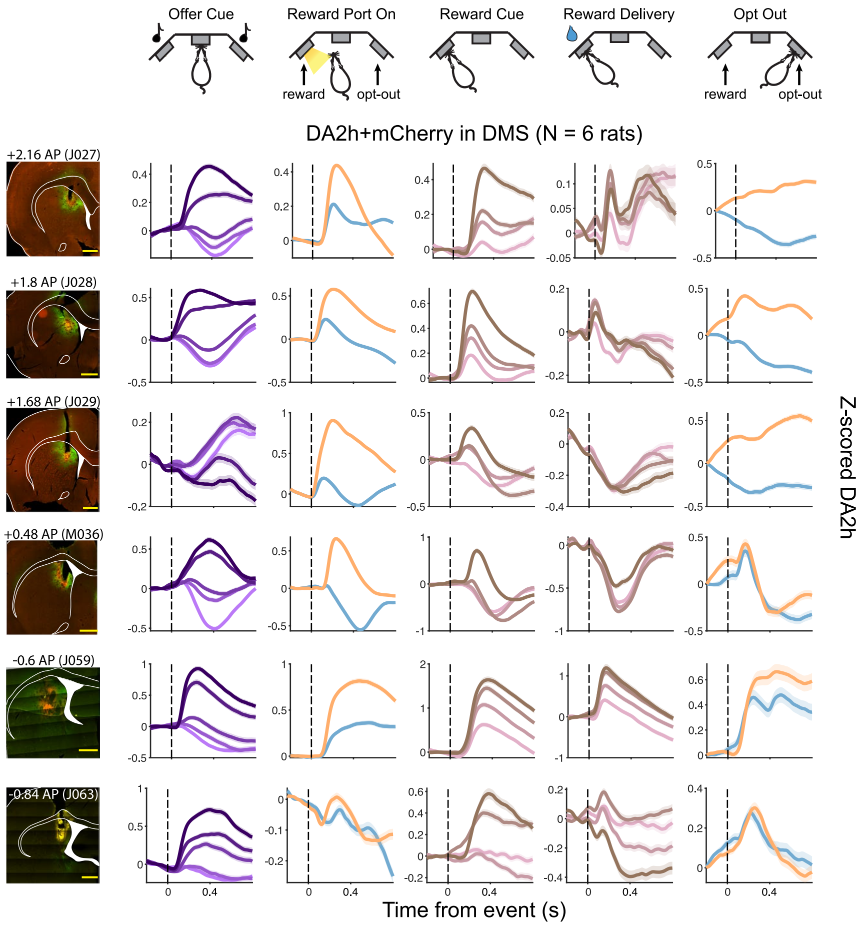

Extended Data Fig. 5 | Characteristic dopamine RPE responses and contralateral selectivity are present across the DMS. Each row shows histology and event-aligned dopamine release for individual rats injected with green-fluorescent DA2h and mCherry (N = 6 rats, mean ± s.e.m. across sessions plotted for each rat). Event-aligned signals are z-scored and baseline corrected. Across panels, the solid yellow line in the histology images indicates a 1-mm scale bar.

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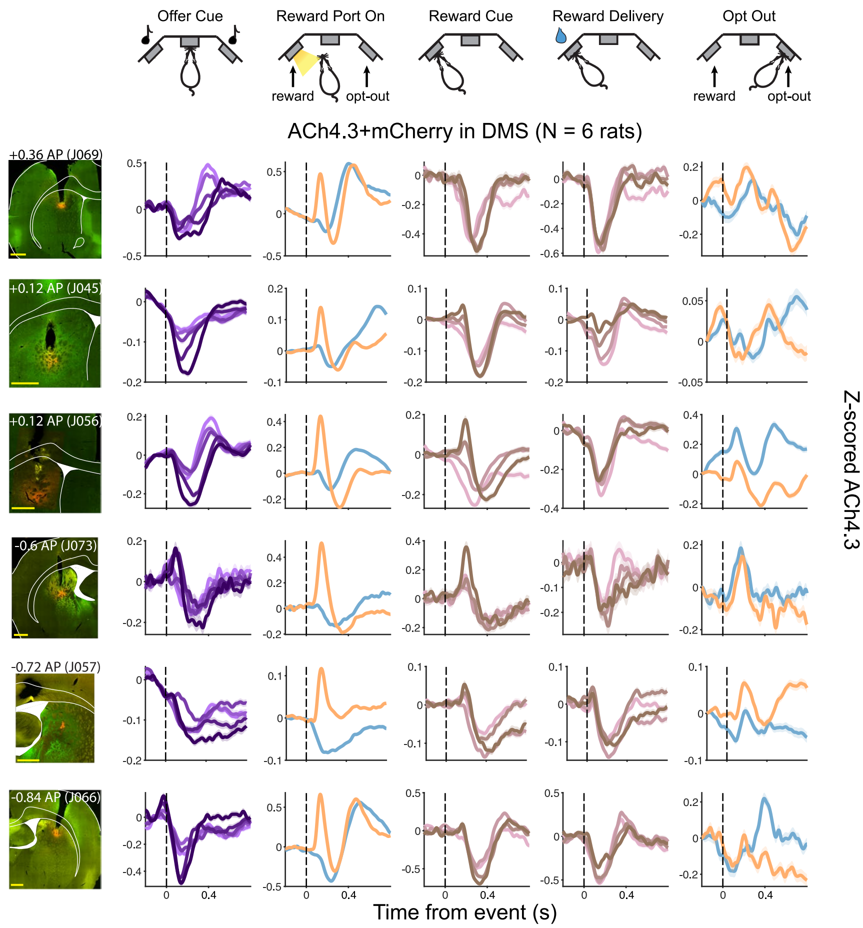

Extended Data Fig. 6 | Characteristic cholinergic dips and bursts at RPE- and movement-associated events are present across the DMS. Each row shows histology and event-aligned acetylcholine release for individual rats injected with green-fluorescent ACh4.3 and mCherry (N = 6 rats, mean ± s.e.m. across sessions plotted for each rat). Event-aligned signals are z-scored and baseline-corrected. Across panels, the solid yellow line in the histology images indicates a 1-mm scale bar.

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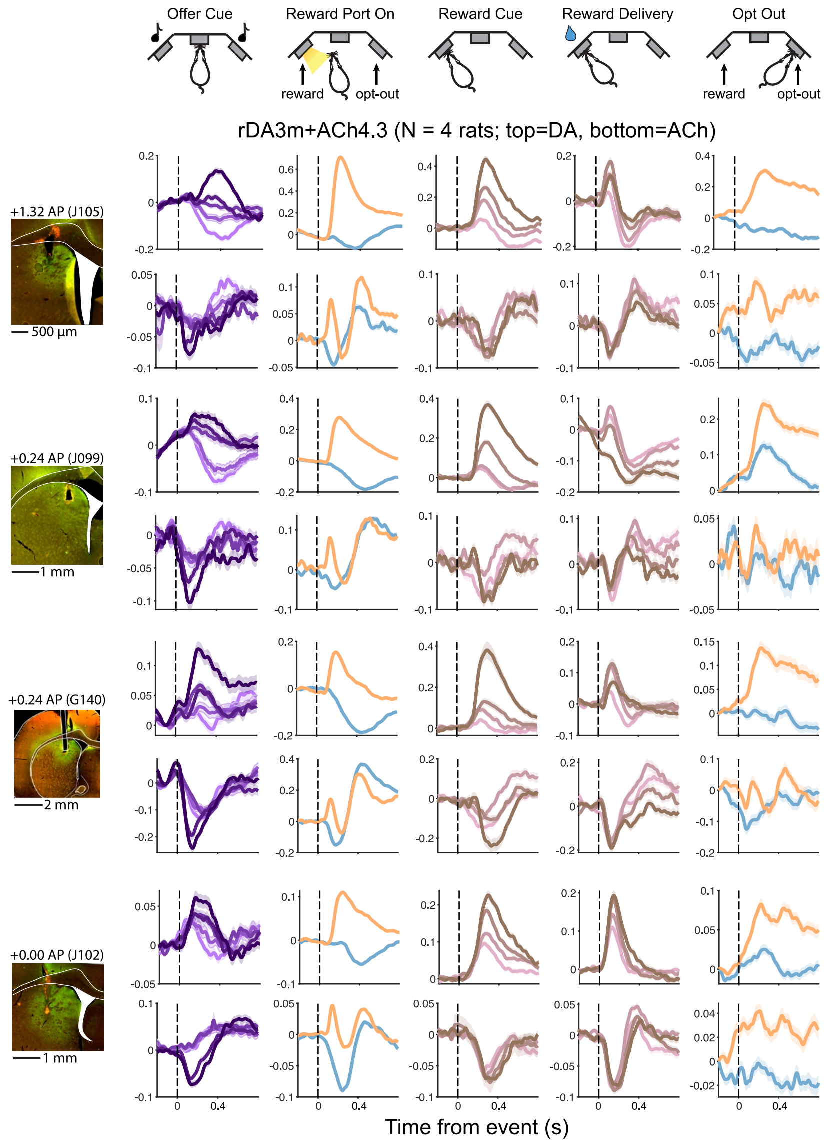

Extended Data Fig. 7 | Simultaneous measurement of dopamine and acetylcholine release reveals characteristic dopamine and acetylcholine dynamics at RPE- and movement-associated events across the DMS. Event-aligned dopamine (top) and acetylcholine (bottom) responses in rats injected with red-shifted rDA3m and green-fluorescent ACh4.3 (N = 4 rats, mean ± s.e.m. across sessions plotted for each rat). Event-aligned signals are z-scored and baseline-corrected.

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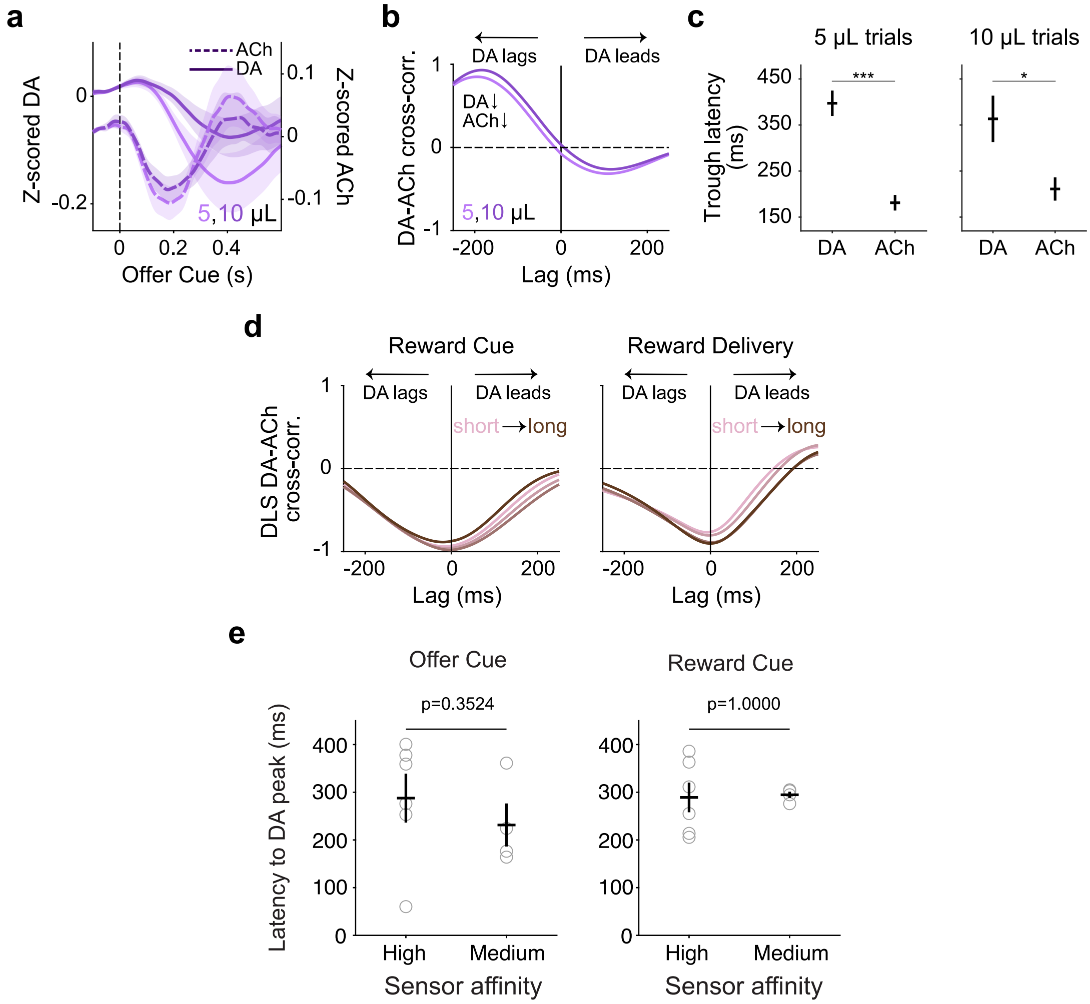
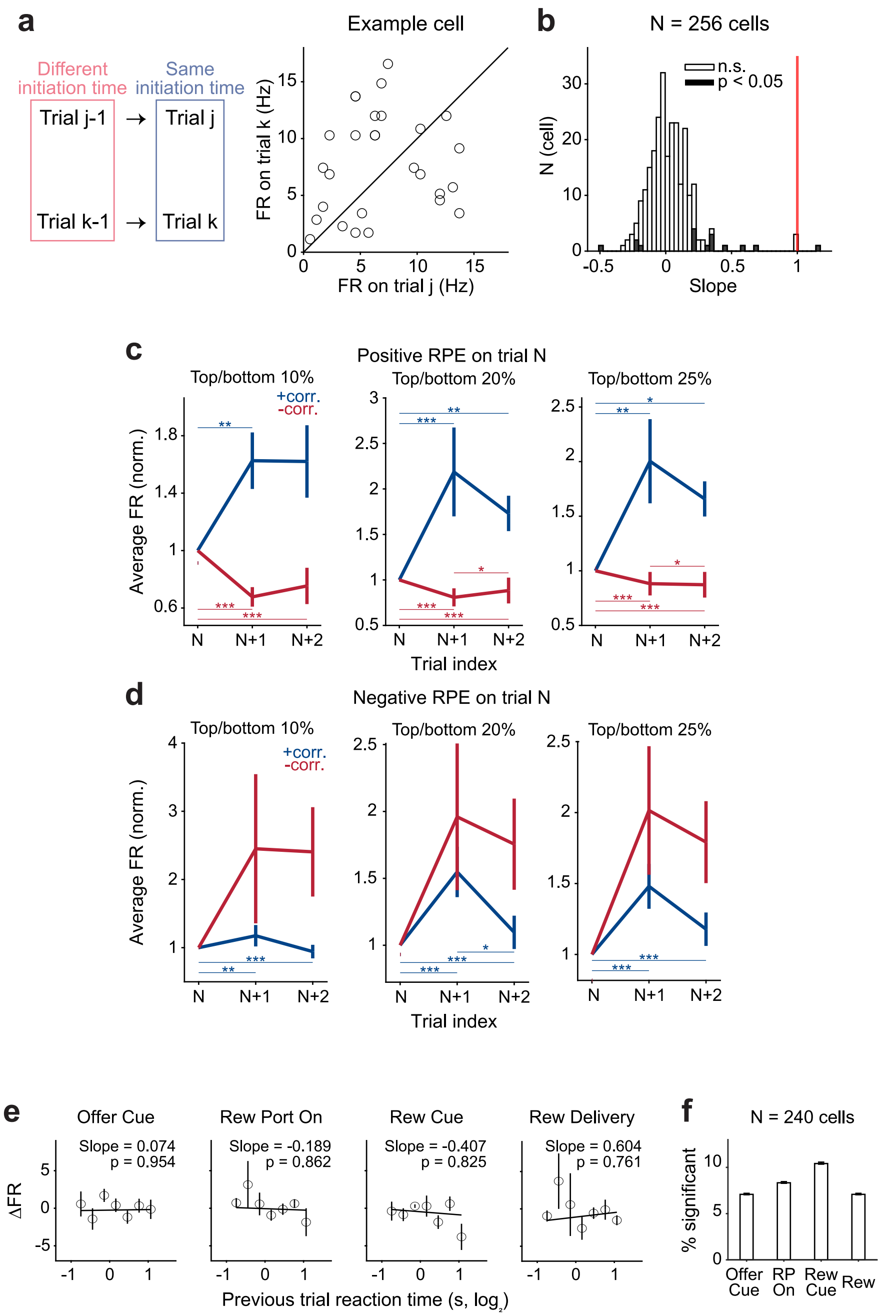


Extended Data Fig. 8 | DMS acetylcholine and dopamine dynamics at the offer cue and DLS acetylcholine and dopamine dynamics at reward. a, Acetylcholine dips slightly lead dips in dopamine at the offer cue. Z-scored DMS dopamine (solid, N = 10 rats, mean ± s.e.m.) and acetylcholine (dashed, N = 10 rats, mean ± s.e.m.) signals for small reward offer trials in mixed blocks aligned to the onset of offer cue, averaged across rats. **b**, Cross-correlation between average dopamine and acetylcholine signals around offer cue (−0.1 to 0.5 s) for small reward offer trials. Negative lag indicates dopamine following acetylcholine. **c**, Latency to trough in DMS dopamine and DMS acetylcholine release after the onset of offer cue on 5 µl (left) and 10 µl (right) offer trials in mixed block, averaged across rats (N = 10 DA rats, N = 10 ACh rats, mean ± s.e.m.; two-sided Wilcoxon rank-sum test: 5-µl trials, $p = 4.35 \times 10^{-4}$; 10-µl trials, p = 0.0140). **d**, Acetylcholine and dopamine dynamics in the DLS are antiphase at the time of reward cue and delivery. Cross-correlation of rat-averaged DLS dopamine and acetylcholine signals (−0.1 to 0.5 s) at the reward cue (left) and reward delivery (right) on trials with different reward delay quartiles. N = 3 rats. **e**, Latency to peak in DMS dopamine at the time of the offer cue (left) and reward cue (right) by sensor affinity, averaged across rats (N = 10 rats for high-affinity, N = 4 rats for medium-affinity, mean ± s.e.m.; two-sided Wilcoxon rank-sum test: offer cue, p = 0.3524; reward cue, p = 1.0000). Gray circles are individual rats.



Extended Data Fig. 9 | Effects of current trial initiation speed and previous trial RPE across a range of RPE thresholds on DMS firing rates. a, Trial initiation speed on the current trial alone does not explain DMS firing rate. Left, trials j and k have similar trial initiation times (difference < 0.01 s) but different initiation times on the previous trial (difference > 2 s). Right, average firing rate at offer cue (0–0.3 s) on trials j and k for an example cell. Solid line is the unity line. **b**, Histogram of best-fit line slopes for all cells. Filled bars indicate cells with significant slopes (two-sided t-test, $p < 0.05$). Red vertical line indicates slope of 1. **c**, RPE produces lasting changes in task-evoked firing rates for a range of RPE thresholds. Change in average z-scored firing rate at offer cue (0–0.3 s) on 3 consecutive trials where a big positive RPE on trial N is followed by a negligible RPE on trial N + 1. Different panels are for different thresholds used to select RPE on trial N. Mean ± s.e.m. across trial sequences, two-sided Wilcoxon signed-rank: left, 10% threshold: +corr. cells, N = 100 trial sequences, trial N vs N + 1, $p = 0.0064$; trial N vs N + 2, $p = 0.0620$; trial N + 1 vs N + 2, $p = 0.4269$; −corr. cells, N = 115 trial sequences, trial N vs N + 1, $p = 8.04 \times 10^{-10}$; trial N vs N + 2, $p = 5.39 \times 10^{-12}$; trial N + 1 vs N + 2, $p = 0.5312$. Middle, 20% threshold: +corr. cells, N = 191 trial sequences, trial N vs N + 1, $p = 1.42 \times 10^{-4}$; trial N vs N + 2, $p = 0.0395$; trial N + 1 vs N + 2, $p = 0.1138$; - corr. cells, N = 223 trial sequences, trial N vs N + 1, $p = 1.44 \times 10^{-13}$; trial N vs N + 2, $p = 4.78 \times 10^{-19}$; trial N + 1 vs N + 2, $p = 0.0182$. Right, 25% threshold: +corr. cells, N = 245 trial sequences, trial N vs trial N + 1, $p = 0.0025$; trial N vs trial N + 2, $p = 0.0434$; trial N + 1 vs trial N + 2, $p = 0.3320$; −corr. cells, N = 280 trial sequences, trial N vs trial N + 1, $p = 5.05 \times 10^{-16}$; trial N vs trial N + 2, $p = 8.38 \times 10^{-21}$; trial N + 1 vs trial N + 2, $p = 0.0339$. **d**, Same as in **c** but for a big negative RPE on trial N. Mean ± s.e.m. across trial sequences, two-sided Wilcoxon signed-rank: left, +corr. cells, N = 119 trial sequences, trial N vs trial N + 1, $p = 9.77 \times 10^{-4}$, trial N vs trial N + 2, $p = 6.51 \times 10^{-6}$; middle, +corr. cells, N = 240 trial sequences, trial N vs trial N + 1, $p = 5.00 \times 10^{-4}$; trial N vs trial N + 2, $p = 2.09 \times 10^{-9}$; trial N+1 vs trial N+2, $p = 0.0108$; Right, +corr. cells, N = 299 trial sequences, trial N vs trial N + 1, $p = 1.92 \times 10^{-5}$; trial N vs trial N + 2, $p = 1.60 \times 10^{-8}$. Across panels, *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$. **e**, Speed of contralateral movement does not modulate DMS firing rate on subsequent trial. Trial-by-trial changes in average firing rate surrounding each task event (0–0.3 s) versus putative DA at reward port on (that is, vigor of contralateral movement) for an example cell. Solid lines show the best-fit line (two-sided t-test on the slope: offer cue, $p = 0.954$; reward port on, $p = 0.862$; reward cue, $p = 0.825$; rew delivery = 0.761). Trials are binned in 7 linearly spaced intervals for visualization (N = 111 trials, mean ± s.e.m.). **f**, Fraction of cells with significant slopes at different task events. Mean ± 95% binomial confidence intervals.

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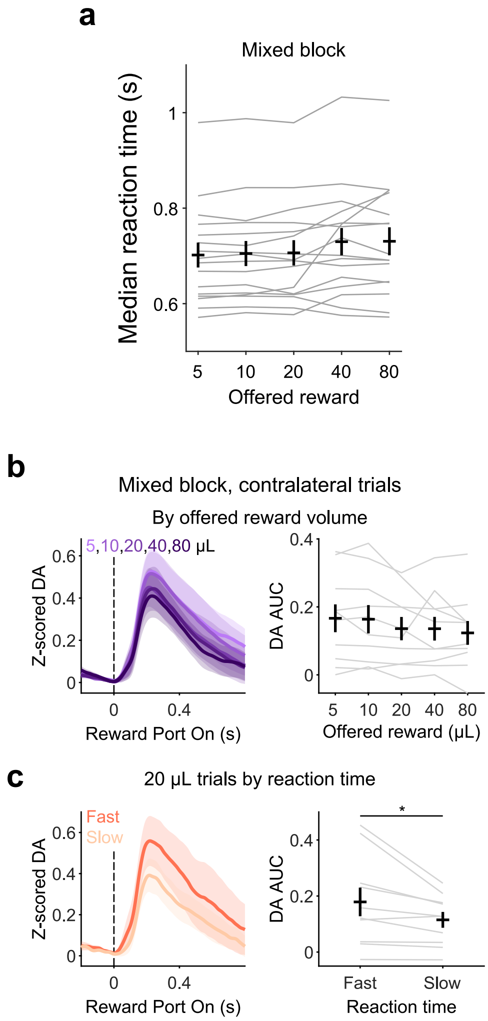

Extended Data Fig. 10 | DMS dopamine predicts vigor of orienting to the reward port, independent of the reward offer. a, Vigor of orienting to the reward port upon illumination does not correlate with the reward offer volume. Median reaction time to the reward port LED illumination at the reward port assignment in mixed blocks for different reward offers, averaged across rats (N = 16 rats, mean ± s.e.m.; linear mixed-effects model fit on median reaction time vs reward volume: slope = 0.0082, two-sided t-test on slope: $p = 4.92 \times 10^{-4}$). Gray lines indicate individual rats. The lack of reward-dependent modulation of movement speed is consistent with this orienting movement reflecting a prepotent, reflexive response to a salient stimulus (the LED turning on). **b**, Dopamine release at reward port assignment is not modulated by value. Dopamine release (left) and AUC (right) for varying reward offer volumes on contralateral trials during mixed blocks, averaged across rats (N = 10 rats, mean ± s.e.m.). Gray lines indicate individual rats. **c**, Dopamine release (left) and AUC (right) for 20-µl contralateral trials during mixed blocks when the side LED orienting response is in the fastest versus slowest quartile (N = 10 rats, mean ± s.e.m.; two-sided Wilcoxon signed-rank: p = 0.0137). Gray lines indicate individual rats.

nature portfolio

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For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed	
☐	☒	The exact sample size (*n*) for each experimental group/condition, given as a discrete number and unit of measurement
☐	☒	A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☐	☒	The statistical test(s) used AND whether they are one- or two-sided *Only common tests should be described solely by name; describe more complex techniques in the Methods section.*
☐	☒	A description of all covariates tested
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☐	☒	A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☐	☒	For null hypothesis testing, the test statistic (e.g. *F*, *t*, *r*) with confidence intervals, effect sizes, degrees of freedom and *P* value noted *Give P values as exact values whenever suitable.*
☒	☐	For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐	For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☒	☐	Estimates of effect sizes (e.g. Cohen's *d*, Pearson's *r*), indicating how they were calculated

Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	Most videos and fiber photometry recordings were collected through Doric Neuroscience Studio version 6 software. Neuropixels data were acquired through OpenEphys versions 0.6.5 or 0.6.6.
Data analysis	DeepLabCut software version 2.2.3 was used to track features from acquired videos for all experiment types. Fibre photometry recordings were motion-corrected using a code from published literature, available on github (https://github.com/Nondairy-Creamer/tmac). Neuropixels data were preprocessed using Kilosort version 2.5 and Phy2 Python package. All other analyses were done using custom Matlab codes (R2021b), to be found on github (https://github.com/constantinoplelab/published/tree/main/DMS_AChDA).

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio guidelines for submitting code & software for further information.

Data

Policy information about availability of data

All manuscripts must include a data availability statement. This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our policy

Data required to generate figures are available on Zenodo. The link is provided in README.md at https://github.com/constantinoplelab/published/tree/main/DMS_AChDA

Research involving human participants, their data, or biological material

Policy information about studies with human participants or human data. See also policy information about sex, gender (identity/presentation), and sexual orientation and race, ethnicity and racism.

Reporting on sex and gender	*Use the terms sex (biological attribute) and gender (shaped by social and cultural circumstances) carefully in order to avoid confusing both terms. Indicate if findings apply to only one sex or gender; describe whether sex and gender were considered in study design; whether sex and/or gender was determined based on self-reporting or assigned and methods used. Provide in the source data disaggregated sex and gender data, where this information has been collected, and if consent has been obtained for sharing of individual-level data; provide overall numbers in this Reporting Summary. Please state if this information has not been collected. Report sex- and gender-based analyses where performed, justify reasons for lack of sex- and gender-based analysis.*
Reporting on race, ethnicity, or other socially relevant groupings	*Please specify the socially constructed or socially relevant categorization variable(s) used in your manuscript and explain why they were used. Please note that such variables should not be used as proxies for other socially constructed/relevant variables (for example, race or ethnicity should not be used as a proxy for socioeconomic status). Provide clear definitions of the relevant terms used, how they were provided (by the participants/respondents, the researchers, or third parties), and the method(s) used to classify people into the different categories (e.g. self-report, census or administrative data, social media data, etc.) Please provide details about how you controlled for confounding variables in your analyses.*
Population characteristics	*Describe the covariate-relevant population characteristics of the human research participants (e.g. age, genotypic information, past and current diagnosis and treatment categories). If you filled out the behavioural & social sciences study design questions and have nothing to add here, write "See above."*
Recruitment	*Describe how participants were recruited. Outline any potential self-selection bias or other biases that may be present and how these are likely to impact results.*
Ethics oversight	*Identify the organization(s) that approved the study protocol.*

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	Statistical calculations were not used to choose sample size. Sample sizes were determined based on sufficient trials/sessions for statistical tests.
Data exclusions	Fiber photometry recording sessions with low signal-to-noise ratio were excluded. Animals with fiber placements outside the brain region of interest were excluded based on histology.
Replication	Statistical tests were used to ensure that the findings are not driven by a small subset of the population. Where applicable, individual data are plotted along with the population data.
Randomization	For fiber photometry experiments, fiber implant hemisphere was randomized.
Blinding	Blinding was not relevant for fiber photometry and Neuropixels experiments.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study	
☐	☒	Antibodies
☒	☐	Eukaryotic cell lines
☒	☐	Palaeontology and archaeology
☐	☒	Animals and other organisms
☒	☐	Clinical data
☒	☐	Dual use research of concern
☒	☐	Plants

Methods

n/a	Involved in the study	
☒	☐	ChIP-seq
☒	☐	Flow cytometry
☒	☐	MRI-based neuroimaging

Antibodies

Antibodies used	Rabbit anti-GFP: Thermo Fisher Scientific A11122, lot #2083201 Goat anti-Rabbit AlexaFluor 488 IgG: Thermo Fisher Scientific A11008, lot #2147635
Validation	GFP: From manufacturer, "Antibody specificity was demonstrated by detection of different targets fused to GFP tag in transiently transfected lysates tested. Relative detection of GFP tag was observed across different proteins fused with GFP in H3-GFP (Lane 3-5) and p65-GFP (Lane 6). GFP-variant, YFP is also being detected in His-p65-YFP lysate (Lane 7), using Anti-GFP Polyclonal Antibody (Product # A11122) in Western Blot." Many previous works have used this antibody, including: Roth et al. "Spatially controlled construction of assembloids using bioprinting." Nature Communications volume 14, (2023) Suzuki et al. "A cortical cell ensemble in the posterior parietal cortex controls past experience-dependent memory updating." Nature communications volume 13 (2022)

Animals and other research organisms

Policy information about studies involving animals; ARRIVE guidelines recommended for reporting animal research, and Sex and Gender in Research

Laboratory animals	79 Long-Evans rats (60 males, 19 females) between ages 6 and 24 months were used for this study (15 males and 7 females for fiber photometry, 8 males for Neuropixels, 41 males and 12 females for behavioral analysis including 4 that also underwent fiber photometry). 3 rats were TH-Cre and 1 rat was Adora2a transgenic.
Wild animals	This study did not involve wild animals.
Reporting on sex	Both male and female rats were used. Previous publication from the lab using identical behaviour and training protocols show no differences in behaviour: Mah et al. "Distinct value computations support rapid sequential decisions." Nature Communications volume 14, (2023)
Field-collected samples	This study did not involve field-collected samples.
Ethics oversight	Animal use procedures were approved by he New York University Animal Welfare Committee (UAWC #2021-1120) and carried out in accordance with National Institute of Health standards.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Plants

Seed stocks	*Report on the source of all seed stocks or other plant material used. If applicable, state the seed stock centre and catalogue number. If plant specimens were collected from the field, describe the collection location, date and sampling procedures.*
Novel plant genotypes	*Describe the methods by which all novel plant genotypes were produced. This includes those generated by transgenic approaches, gene editing, chemical/radiation-based mutagenesis and hybridization. For transgenic lines, describe the transformation method, the number of independent lines analyzed and the generation upon which experiments were performed. For gene-edited lines, describe the editor used, the endogenous sequence targeted for editing, the targeting guide RNA sequence (if applicable) and how the editor was applied.*
Authentication	*Describe any authentication procedures for each seed stock used or novel genotype generated. Describe any experiments used to assess the effect of a mutation and, where applicable, how potential secondary effects (e.g. second site T-DNA insertions, mosiacism, off-target gene editing) were examined.*